A General Capacity Frontier of Complex Networks

Nicholas Kunz* nhk37@cornell.edu
H. Oliver Gao hg55@cornell.edu
*Corresponding Author**

Systems Engineering, Cornell University

October 12, 2026

Abstract

From cells to cities, ecosystems to economies, complex networks exhibit peak intensities that define their empirical limit. Estimating these limits traditionally requires domain-specific knowledge. We demonstrate that a wide range of complex networks share a capacity frontier that constrains these maxima. We formalize this regularity by proposing Structural Capacity Theory for Network Substrate Structured Systems. These systems couple a fixed network substrate with an observable dynamical process. The frontier is learned from a vector representation of the network substrate and modulated log-additively by a vector representation of the dynamical process. Frontiers trained on one set of domains reliably bound maxima of entirely different ones without observing their peak intensities or simulating their dynamics. Independent learning paradigms converge on consistent upper bounds. Frontier performance is stable under perturbation, yet degrades under deliberate falsification, with no performance gain under ablation. These findings support the capacity frontier as a generalizable constraint that describes how network structure bounds dynamical processes across otherwise unrelated systems.

Maxima are fundamental features of complex networks. Understanding them is important because they often mark where critical thresholds are met that cause material changes in network behavior [1–5]. For example, a regulatory network that exceeds its threshold of localized molecular activity changes regimes [2, 6]. A transportation network that exceeds its carrying capacity changes behavior primarily by congestion, not by design [7, 8]. Although it is known that network maxima can affect network behavior, understanding *how* traditionally requires domain-specific knowledge [2, 7]. Whether or not network maxima have a common regularity across domains is still an open question [9–11]. Addressing it requires a shared basis of comparison.

In the previous examples, network structure formed the basis on which a dynamical process occurred [2, 4, 7, 8, 11, 12]. Size, connectivity, spectral organization, and other topological characteristics constrain network throughput [13–16], propagation [17–19], synchronization [20], controllability [21], and spatial transport [22]. The same network substrate can have different limits depending on the observational definition, activity attribution, temporal aggregation, and measurement regime [23–25]. Conversely, structurally distinct substrates and dynamical processes can exhibit comparable limits under shared measurement conventions and operational definitions [11, 26–28]. In both cases, maxima reflect the network substrate and the dynamical process by which observations are attributed, generated, and measured.

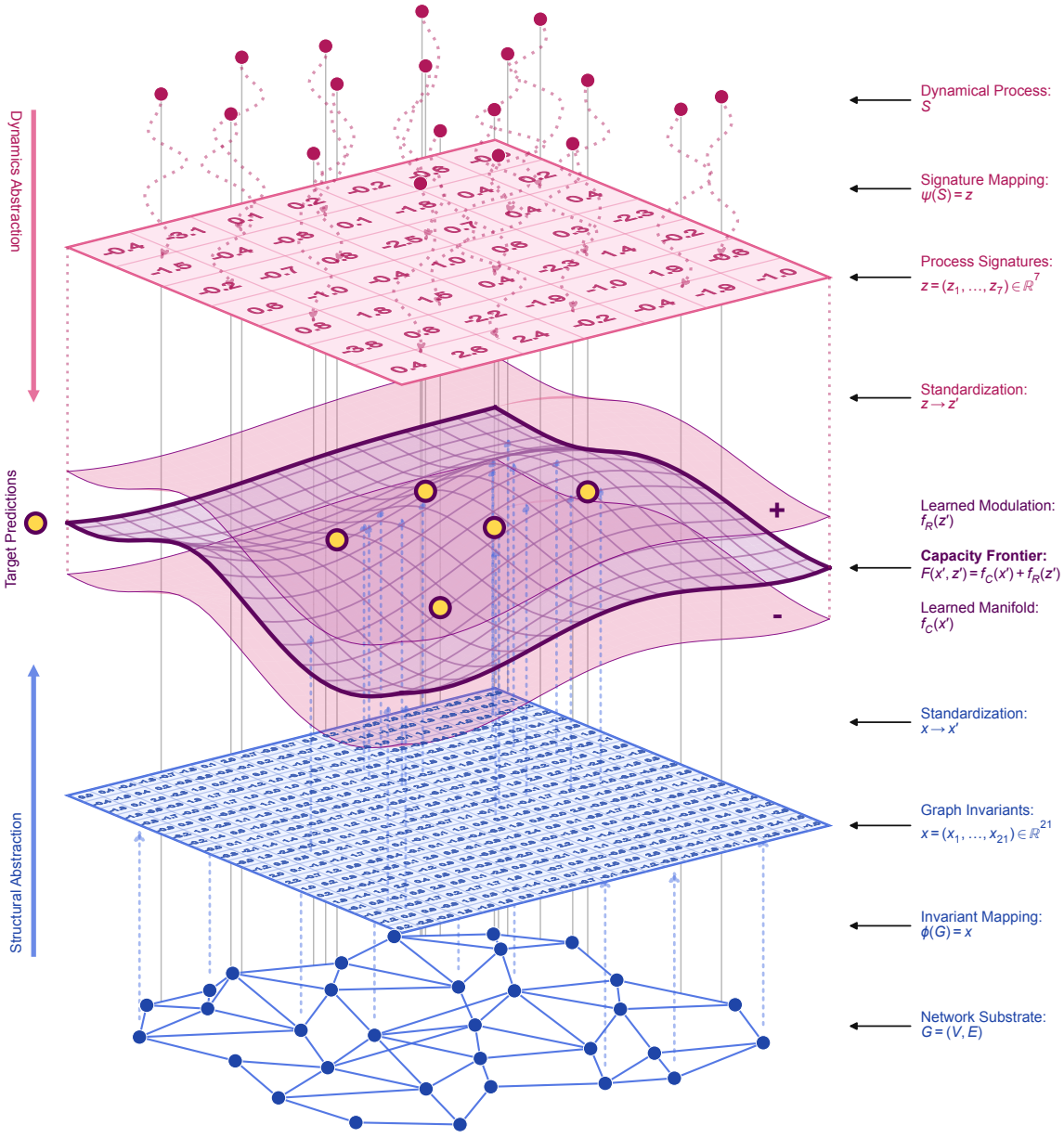


Figure 1: **Conceptual representation of Structural Capacity Theory (SCT).** Schematic illustration of Network Substrate Structured Systems (N3S). The network substrate G (blue graph, bottom) is encoded as a standardized vector x' on the structural plane. The dynamical process S (pink markers, top) produces observations attributed to network substrate elements and encoded as a standardized vector z' on the dynamics plane. The curved manifold (purple surface, middle) represents the capacity frontier $F(x', z') = f_C(x') + f_R(z')$, where the learned structural component $f_C(x')$ is modulated log-additively by the learned process component $f_R(z')$ to estimate capacity frontier values $\widehat{y}^*$ (gold points, middle).

We propose Network Substrate Structured Systems (N3S) as a class of systems for which this coupling can be evaluated. An N3S is defined by a fixed network substrate, an observable dynamical process that corresponds to that substrate, and a measurable maximum rate that corresponds to that dynamical process (Fig. 1). Membership is determined from the observations a given system naturally provides so that the scope of the theory is fixed before it is tested. Structural Capacity Theory (SCT) proposes that maxima from any N3S are comparable when represented in a shared coordinate system, and within that representation, maxima are constrained by a common capacity frontier.

We estimate the capacity frontier as a covariate-dependent upper-envelope [29, 30]. Typical network activity can be captured by central tendencies, but the area of interest is the ceiling, so the estimators are trained as such [31–37]. The system's network substrate G and dynamical process S are encoded as graph invariants x and process signatures z, and standardized to x' and z', respectively. The capacity frontier $F(x', z') = f_C(x') + f_R(z')$ (Eq. (7)) log-additively bounds the log-transformed maximum rate $y^* = \ln(1 + r_{\max})$ (Eq. (3)). The frontier is strictly empirical. It permits a small fraction of observations to cross it and carries no claim of analytical containment.

We evaluate whether this formulation supports a transferable, consistent, stable, falsifiable, and decomposable capacity frontier. We found a frontier that bounded maxima from out-of-sample systems using Leave-One-Group-Out Cross-Validation (LOGO-CV) and repeated k-Fold Cross-Validation (k-Fold-CV) with equivalent performance. Independent estimators from 3 distinct learning paradigms predicted consistent frontier values. Perturbations to network substrates, graph invariants, dynamical processes, and process signatures preserved frontier performance. By contrast, 3 falsification procedures that deliberately destroyed the representation–target correspondence degraded frontier performance, which retraining could not recover. Finally, ablations that tested simpler and more complex alternative formulations did not meaningfully improve the performance of the frontier.

Shared Coordinate System

Encoding all systems as paired network substrate and dynamical process vectors placed unrelated phenomena within a shared coordinate system (Fig. 1; Supplementary Table 1). The corpus of 25 systems contained 5 broad domains: Earth & Physical Sciences, Life Sciences & Medicine, Technology & Information, Trade & Institutions, and Transportation & Infrastructure. Each of the 5 broad domains contained 5 individual disciplines. Their measurements did not share units before encoding. The network substrate vector x admitted 21 graph invariants (Supplementary Table 7). The dynamical process vector z admitted 7 process signatures (Supplementary Table 8). Both vectors were selected under prespecified criteria and standardized to x' and z', respectively (Encoding and Standardization). This encoding recast every system with the same number of dimensions despite large differences in substance and scale. It made the diverse corpus of 25 systems directly comparable.

Within this shared representation, the capacity frontier bounded maxima across the corpus (Supplementary Table 2). Under repeated k-Fold-CV, the frontier was fit on each training set and evaluated on the corresponding test set (Resampling and Leakage Control). Each test was repeated 30 times to reduce dependence on any single seeded fold. Frontier performance was reported as medians across estimators after averaging them over test folds and repetitions. Under repeated 5-Fold-CV and repeated 10-Fold-CV, frontier efficiency was 0.77 and 0.76, respectively. Frontier efficiency was measured using the Efficiency Index (EI) — the bounded $[0, 1]$ geometric mean of the non-violation rate and two bounded metrics that decreased with violation magnitude and slack relative to the evaluated target scale (Performance Metrics, Eq. (8)). There were infrequent frontier violations, low violation magnitude, and consistent slack. Under repeated 5-Fold-CV, the violation rate was 0.02, mean violation magnitude 0.06, and the mean slack 6.86 log units. Under repeated 10-Fold-CV, the corresponding values were 0.01, 0.02, and 6.87 log units, respectively. Thus, the shared representation supported a common capacity frontier.

Transfer Across Domains

A capacity frontier trained without a given domain constrained the maxima of that domain's disciplines (Supplementary Table 2). Under domain-level LOGO-CV, the frontier was trained on 4 of the 5 broad domains and evaluated on the 5 individual disciplines in the withheld domain. Frontier efficiency was 0.75, ranging from 0.70 to 0.91 (Fig. 2a). The violation rate was 0.04, mean violation magnitude 0.08, and mean slack 6.86 log units. Estimation-observation consensus was 0.84, ranging from 0.77 to 0.92 across the same evaluations (Fig. 2b). Estimation-observation consensus was measured using the Consensus Index (CI) — the bounded $[0, 1]$ geometric mean of global rank agreement, high-capacity overlap, and non-linear dependence (Performance Metrics, Eq. (9)). Each withheld discipline supplied its own graph invariants, process signatures, and maximum rate. It tested domain transfer by out-of-sample training, not prediction in the absence of observation. This established generalization across the tested domains.

A capacity frontier trained without a given discipline also bounded that discipline's maxima. Withholding a single discipline tested generalization similarly to domain-level LOGO-CV, but at a higher resolution. Under discipline-level LOGO-CV the frontier was trained on 24 of the 25 disciplines and evaluated on the one that was withheld, effectively Leave-One-Out Cross-Validation (LOO-CV). In this resampling regime frontier efficiency was 0.73, ranging from 0.52 to 0.90 across disciplines. The violation rate was 0.00, mean violation magnitude 0.00, and mean slack 6.12 log units. Estimation-observation consensus was 0.82 based on out-of-sample predictions pooled within each of the 5 broad domains, ranging from 0.77 to 0.92 across domains. Violations were less common and smaller than under domain-level LOGO-CV. The experiment could not determine whether these differences reflected denser coverage of the coordinate system by in-sample disciplines or differences in test set size and metric aggregation. Nevertheless, the frontier generalized to out-of-sample disciplines similarly to out-of-sample domains.

The capacity frontier achieved equivalent frontier performance across resampling regimes. Using the repeated k-Fold-CV results (Shared Coordinate System), paired differences in frontier performance between domain-level LOGO-CV and repeated k-Fold-CV were tested across disciplines within the 5 broad domains (Fig. 2c). Paired differences were computed by subtracting domain-level LOGO-CV results from those contained in the comparative resampling regime, where negative values indicated lower performance. Median paired differences in frontier efficiency were 0.00 for all resampling regimes. Median paired differences in estimation-observation consensus were at most 0.01 in magnitude under the same evaluation. Paired Wilcoxon Two One-Sided Tests (TOST) across 45 estimator-by-domain comparisons supported equivalence (Evaluation and Statistical Inference) within their empirically derived margins ($\delta_{\mathrm{EI}} = 0.02$ and $\delta_{\mathrm{CI}} = 0.11$, all Holm-adjusted $p < 0.001$; Supplementary Table 3). Withholding broad domains and individual disciplines preserved frontier performance when compared to repeated random resampling. Therefore, domain generalization did not depend on any single tested resampling regime.

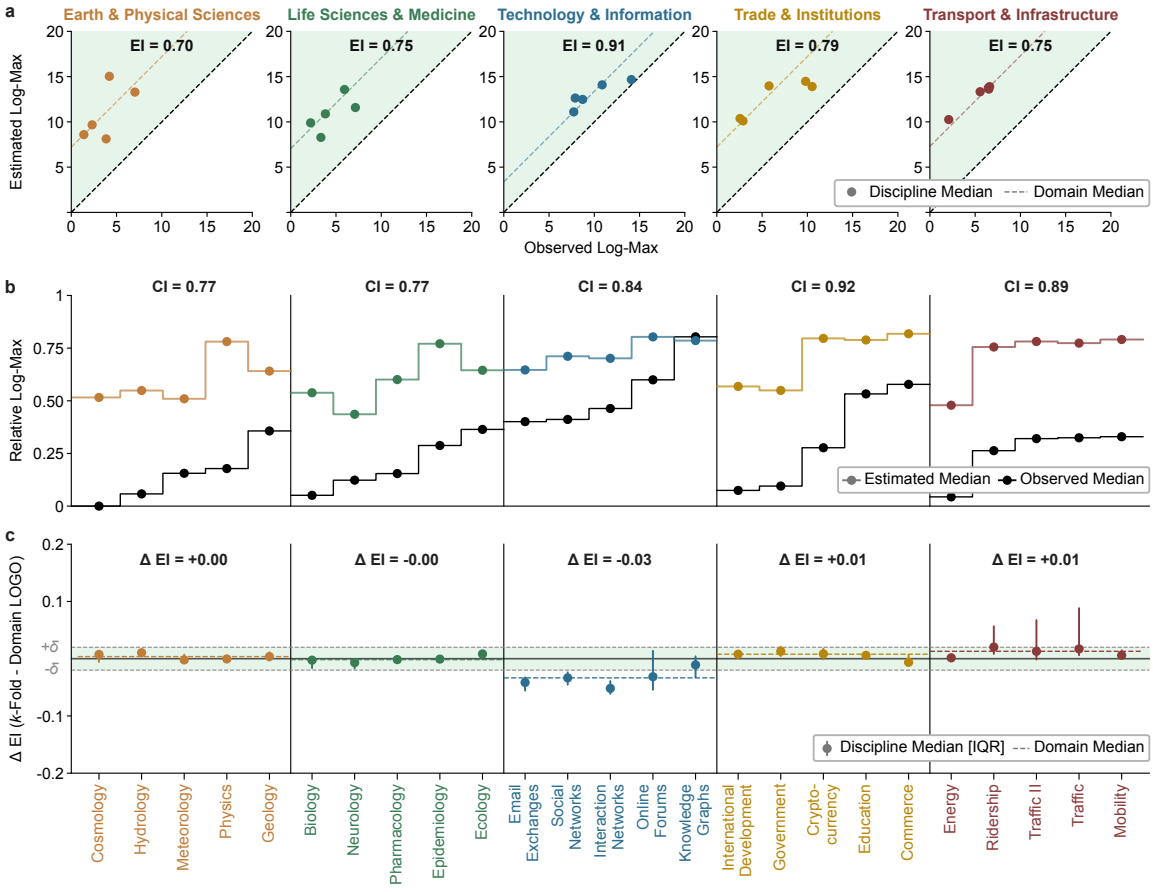


Figure 2: **Generalization of the capacity frontier.** Domain-level LOGO-CV across 25 systems in 5 broad domains. **a**. Estimated versus observed log-maxima for each withheld domain. Discipline medians (domain-color markers) fall along domain offsets (domain-color dashed lines) relative to equality (diagonal gray dashed line). Green shading illustrates estimates at or above log-maxima. **b**. Estimation-observation consensus profiles within withheld domains. Step traces and markers exhibit observed log-maxima (black trace, bottom) and median model estimates (domain-color trace, top), with disciplines ordered by observed maximum and profiles scaled to the unit interval. **c**. Paired differences in frontier efficiency between repeated k-Fold-CV and domain-level LOGO-CV. Markers with vertical intervals exhibit discipline medians and interquartile ranges across models (domain-color intervals), dashed horizontal lines exhibit domain medians (domain-color dashed lines), and the horizontal shading is the equivalence margin ($\pm\delta_{\text{EI}}$, green shaded band). Annotations report (**a**) domain EI, (**b**) domain CI, and (**c**) domain ΔEI.

Consensus Across Paradigms

The capacity frontier was consistent across learning paradigms (Fig. 3). There were 9 independent estimators across 3 distinct learning paradigms, all of which broadly agreed on their predicted frontier values. The learning paradigms were linear parametric, non-linear ensemble, and neural networks (Frontier Learning). All models were trained on the same vectors x' and z' while the estimators differed in functional form, regularization, and optimization. Each test was repeated 30 times to reduce sensitivity to parameter initialization and optimization stochasticity. Predicted frontier values

were averaged across repetitions. Pairwise model consensus across all 36 pairs of seed-averaged predictions was 0.90 (Fig. 3a). A geometric mean this close to 1 simultaneously required high rank agreement, top-weighted overlap, and statistical dependence, such that no single component could carry the outcome. Rescaled Spearman rank correlation (ρ_s^+), Rank-Biased Overlap (RBO), and distance correlation (dCor) were 0.95, 0.86, and 0.91, respectively. The agreement was evaluated on in-sample analyses using full-corpus fits. Consensus across the tested learning paradigms indicated that the frontier was unlikely an artifact of inductive bias (Fig. 3b).

Stable Under Perturbation

The capacity frontier was stable under perturbation (Fig. 4a, Fig. 4b; Supplementary Tables 4 and 9). There were 12 perturbation procedures that introduced controlled modifications to the vectors x' and z' while preserving the target y^* across 4 categories: network substrates, graph invariants, dynamical processes, and process signatures (Perturbation, Falsification, Ablation). Under domain-level LOGO-CV, the frontier was trained on the original data, the model frozen, then evaluated on the perturbed data in the withheld domain. Each test was repeated 30 times and averaged across repetitions to reduce sensitivity to perturbation realization and estimator initialization. Paired differences were computed by subtracting the original results from the perturbed ones across 45 estimator-by-domain comparisons, where negative values indicated lower performance. At the highest tested intensity, median paired differences in frontier efficiency ranged from -0.02 to +0.01. Median paired differences in estimation-observation consensus were 0.00 for all network and invariant perturbations. Median paired differences in consensus were -0.11 for process bootstrapping, -0.02 for process scaling, and -0.06 for process smoothing. Signature jitter, noise, and subset produced median consensus differences of -0.01, -0.01, and -0.04, respectively. Paired Wilcoxon TOST across 45 estimator-by-domain comparisons supported equivalence for all 12 perturbations within their empirically derived margins ($\delta_{\text{EI}} = 0.08$ and $\delta_{\text{CI}} = 0.14$, all Holm-adjusted $p \leq 0.028$; Supplementary Table 4). Response curves across normalized perturbation intensities were descriptive, where only the highest perturbation intensity was formally tested. Nevertheless, the frontier could withstand structural modifications, process distortions, and feature noise while remaining reliable.

Degradation Under Falsification

The capacity frontier predictably degraded under falsification (Fig. 4c; Supplementary Tables 5 and 9). There were 3 falsification procedures that deliberately destroyed the correspondence between the vectors x' and z' and the target y^* in distinct ways: target permutation, random generation, and vector generation (Perturbation, Falsification, Ablation). Under domain-level LOGO-CV, the frontier was evaluated by freezing the models after training on the original data, and separately after retraining on the falsified data. Each test was repeated 30 times and averaged across repetitions to reduce sensitivity to estimator initialization and training stochasticity. Paired differences were computed by subtracting the original results from the falsified ones across 45 estimator-by-domain comparisons, where negative values indicated lower performance. Under the frozen model, median paired differences in frontier efficiency were -0.02 for target permutation, -0.04 for random generation, and -0.03 for vector generation. After retraining, median paired differences in frontier efficiency were -0.09, -0.02, and -0.03, respectively. Median paired differences in estimation-observation consensus were -0.29, -0.28, and -0.21 when frozen, and -0.24, -0.18, and -0.31 after retraining, respectively. Paired One-Sided Wilcoxon Signed-Rank Tests supported lower efficiency and lower consensus across all 12 comparisons (all Holm-adjusted $p \leq 0.013$ for EI and $p < 0.001$ for CI; Supplementary Table 5). The falsified corpora served as negative controls that indicated the frontier depended on a true representation–target correspondence.

Parsimony Under Ablation

No alternative capacity frontier formulation meaningfully outperformed the log-additive decomposition (Fig. 4d; Supplementary Tables 6 and 9). The 4 alternative specifications contained simple ablations: invariants only and signatures only, and complex ablations: joint features and interaction terms (Perturbation, Falsification, Ablation). Invariants only trained the frontier as $F(x')$ alone, while signatures only used $F(z')$ alone. Joint features concatenated both vectors into a single-stage fit $F([x', z'])$, while interaction terms augmented the log-additive decomposition as $F(x', z', h')$ with pairwise products $x_j z_k$. Under domain-level LOGO-CV, ablations were fit on each training set and evaluated on the corresponding test set. Each test was repeated 30 times and averaged across repetitions to reduce sensitivity to estimator initialization and training stochasticity. Paired differences were computed by subtracting the original results from the ablated ones across 45 estimator-by-domain comparisons, where negative values indicated lower performance. Median paired differences in frontier efficiency were -0.02 for joint features, and -0.01 for invariants only, signatures only, and interaction terms. Median paired differences in estimation-observation consensus were -0.20 for invariants only, -0.08 for signatures only, -0.27 for joint features, and -0.10 for interaction terms. Paired One-Sided Wilcoxon Signed-Rank tests supported non-inferiority of the log-additive decomposition to all 4 alternatives in both metrics within their empirically derived margins ($\delta_{\text{EI}} = 0.08$ and $\delta_{\text{CI}} = 0.14$, all Holm-adjusted $p < 0.001$; Supplementary Table 6). These tests limited alternatives and supported the log-additive decomposition as the parsimonious formulation of the frontier.

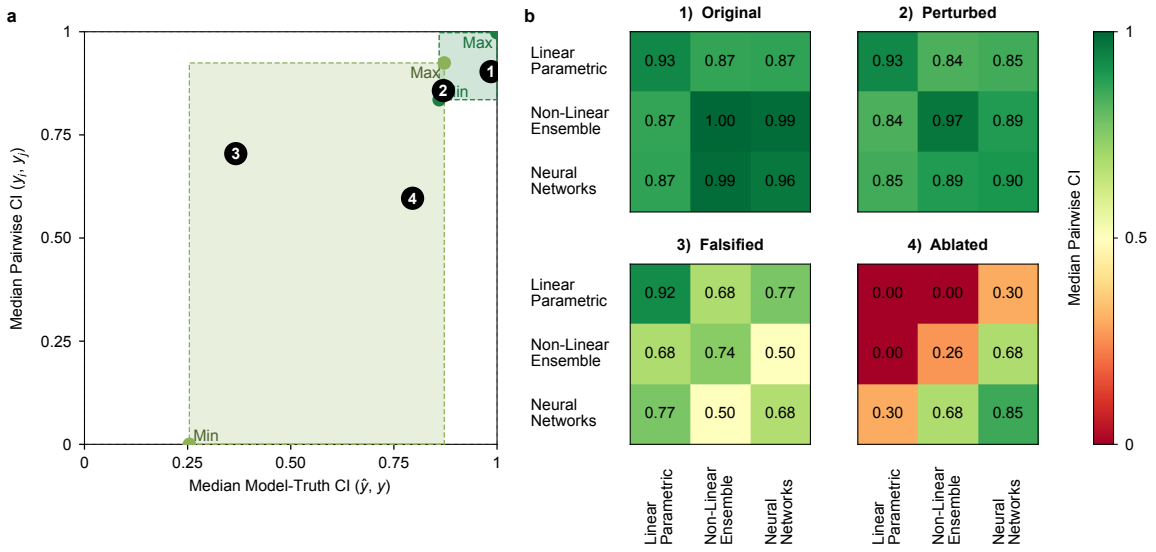


Figure 3: **Learning paradigm and estimation-observation consensus.** 9 estimators across 3 distinct learning paradigms agree on full-corpus fits. **a**. Median estimation-observation consensus ($\widehat{y}^*$ versus y^*) by median pairwise model consensus ($\widehat{y}_i^*$ versus $\widehat{y}_j^*$). Numbered points (black markers, white text) label the tests: (1) original, (2) perturbed, (3) falsified, and (4) ablated. Shaded rectangles span coordinate-wise ranges of learning paradigm medians for the original and perturbed conditions (dashed polygon, green) and falsified and ablated conditions (dashed polygon, yellow). **b**. Median pairwise model consensus across learning paradigms for each test in **a** with paradigm agreement along the diagonal axes.

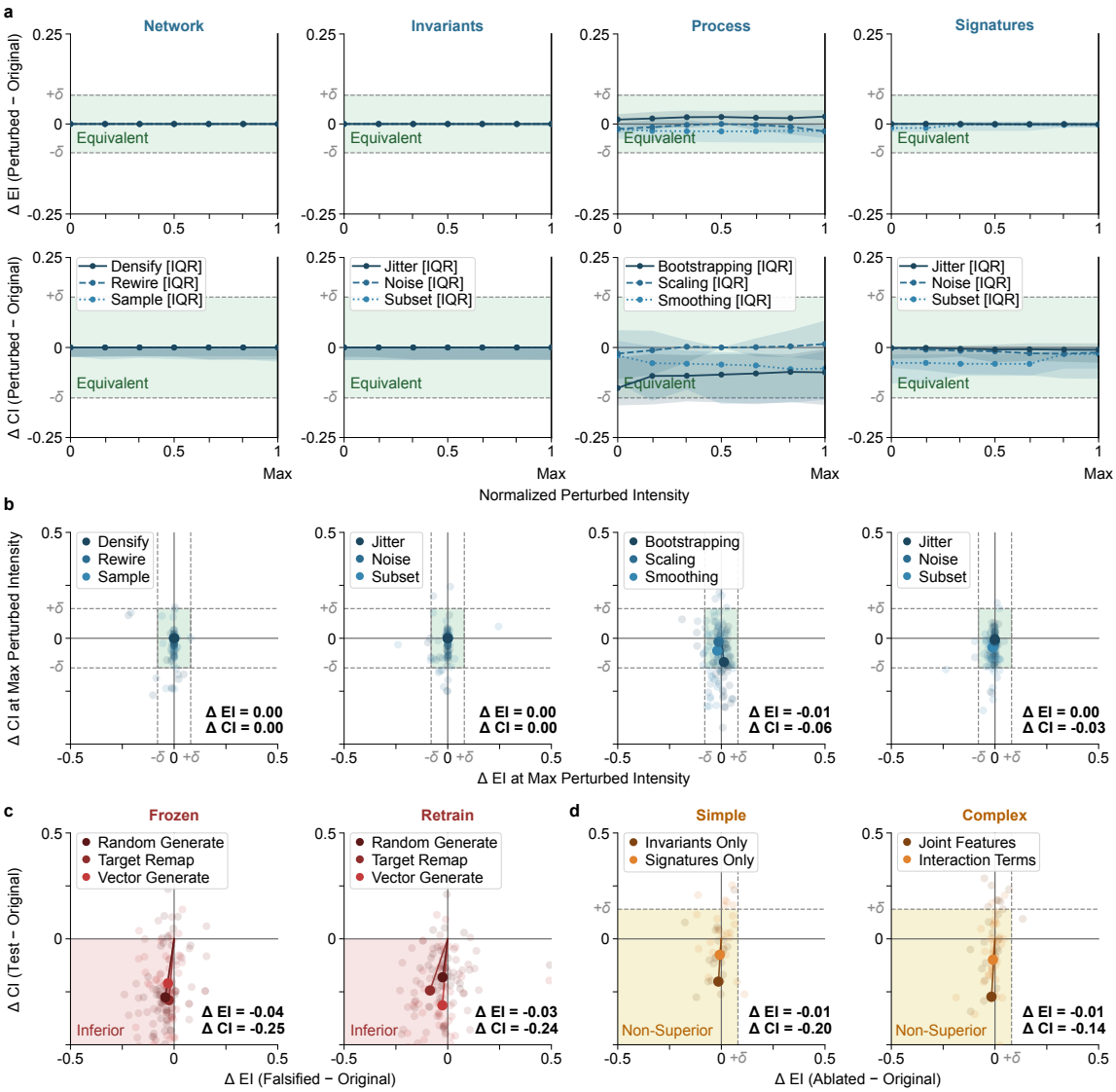


Figure 4: **Stress tests of the capacity frontier.** Paired shifts across 45 estimator-by-domain comparisons under domain-level LOGO-CV, computed as test condition minus original condition or alternative formulations minus log-additive decomposition. **a**. Response curves across normalized intensity for 12 perturbations in 4 categories with ΔEI above and ΔCI below. Blue lines and shaded bands exhibit medians and IQR, respectively. **b**. Paired shifts at highest tested perturbation intensity. **c**. Falsification shifts under frozen and retrain frontiers for 3 falsification procedures. The red quadrant marks joint degradation ($\Delta\text{EI} < 0$, $\Delta\text{CI} < 0$). **d**. Simple (Invariants Only, Signatures Only) and Complex (Joint Features, Interaction Terms) alternative formulations compared with the log-additive decomposition. In **b**–**d**, faint points exhibit individual comparisons and solid points exhibit coordinate-wise medians connected to the origin. Green regions (**a**, **b**) mark perturbation equivalence margins ($\delta_{\text{EI}} = 0.08$, $\delta_{\text{CI}} = 0.14$), and the yellow region (**d**) marks the ablation non-inferiority margin ($\delta_{\text{EI}} = 0.08$, $\delta_{\text{CI}} = 0.14$).

Discussion

The capacity frontier constrains maxima across disciplinary boundaries. A frontier with either a broad domain or individual discipline withheld from training bounded maximum rates with performance comparable to repeated random resampling. Information from one system was therefore useful for estimating maxima in another despite differences in subject, size, scale, units, and dynamics. These comparisons provide a basis for asking which constraints are shared across systems and which depend on domain specificity. Capacity estimation can therefore use empirical evidence beyond a system's own substantive boundary.

Furthermore, the capacity frontier does not require a model of the dynamical processes. Graph invariants and process signatures support capacity estimation without training domain-specific models or simulating their dynamics. Domain-specific models are still important for causal explanation and intervention, but SCT provides a standard benchmark as a complement. Interpreting this benchmark requires additional inquiry, as the reported slack was nearly 3 orders of magnitude higher than the maximum rate. This gap may reflect a loose envelope, unequal observation windows, differences in measurements, or a combination of them. Although the meaning of the slack is still unclear, the frontier still reliably bounds maxima without a traditional model of the underlying dynamics.

Selective stability demonstrates why the capacity frontier is informative. Perturbations at the highest tested intensity generally preserved the frontier's original performance. Falsifications that deliberately destroyed the representation–target correspondence degraded frontier performance, which was unrecoverable even after retraining. Consensus among learning paradigms was relatively high and comparatively similar between the original and perturbed conditions. Only after falsification and ablation did the model consensus between learning paradigms degrade. This supports the frontier as a non-spurious object, distinguishable from a fragile upper-envelope.

The capacity frontier is jointly determined and decomposable. Removing either component or combining them without log-additivity did not improve frontier performance or consensus among estimators. The results support the log-additive decomposition, but leave its exhaustiveness unresolved. All findings were conditioned on the 25 tested systems, 21 graph invariants, 7 process signatures, 9 learned estimators, the log-additive decomposition, and the daily temporal aggregation. Broader corpora, different vector representations, more diverse learning paradigms, and alternative temporal scales all provide direct tests of SCT.

Methods

N3S Definition

Within SCT, an N3S contains a fixed network substrate G, an observable dynamical process S, and a measurable maximum rate $r_{\max}$. The network substrate contains finite elements with relational or attribution structure. The dynamical process is a time-indexed sequence of observed events or reconstructed events from published schedules. Each event is attributable or relates to one or more elements of the network substrate. A given measurement convention converts the sequence into a rate series. Membership requires records for all 3 components. Operational recastings are fixed across all reported evaluations. Systems without an identifiable substrate or with only aggregate observations lacking substrate attribution are excluded from analyses (Supplementary Table 1).

Data and Corpus Composition

The corpus contained 25 systems across 5 broad domains: Earth & Physical Sciences, Life Sciences & Medicine, Technology & Information, Trade & Institutions, and Transportation & Infrastructure. Each of the 5 broad domains contained 5 individual disciplines (Supplementary Table 1; Supplementary Note 2). Network substrates ranged from 5 to 10,356,390 nodes and from 10 to 7,220,486,175 edges. Maximum rates ranged from 3 to 1,323,743 observations per day. Observation windows ranged from 30 to 8,014 days. Measurements shared no units prior to encoding. Each system contributed exactly one record containing its graph invariants x, process signatures z, and maximum rate $r_{\max}$.

Network Substrate Construction

Network substrates were represented as static, unweighted, and undirected graphs. They were strictly held fixed across their observation windows. Nodes represented units. Edges represented native connections between nodes or constructed associations. Directed and weighted edges were converted to undirected and unweighted edges. Multi-edges and self-loops were removed. Isolated nodes were retained. Each network substrate was formally defined as a simple graph G:

$$G \;=\; (V, E), \qquad E \subseteq \binom{V}{2} \tag{1}$$

Let $\mathcal{O}$ denote the observations from which the dynamical process S was constructed. An attribution map assigned each observation $o \in \mathcal{O}$ a nonempty set $a(o) \subseteq V \cup E$, where V and E denote the node and edge sets, respectively. Interaction logs formed unions of observed pairs or complete bipartite graphs over documented entity types. There were 4 network substrates (college, email, mooc, overflow) defined as complete bipartite graphs $K_{m,n}$ by sizes m and n (Supplementary Note 1). Each G served as the fixed network substrate to which observations of the system's dynamical process were attributed.

Dynamical Process Construction

Dynamical processes were constructed from empirical observations. For continuous records, measurements were discretized using system-specific thresholds (Supplementary Table 1). Event counts were aggregated into uniform daily intervals using UTC timestamps or consecutive 24-hour periods. Note the exception of 1 system (celegans) where developmental transitions were reconstructed from published timings using a 30-day dauer duration. Each dynamical process was formally defined as a chronologically ordered discrete daily rate series S:

$$S \;=\; \{r_t\}_{t\in T}\,, \qquad r_t \;=\; \frac{n_t}{\Delta t}, \qquad \Delta t \;=\; 1 \text{ day (24 hours)} \tag{2}$$

where n_t is the event count in interval t and T is the chronologically ordered set of intervals. The number of included intervals was $L = |T|$, with series mean $\bar{r} = L^{-1}\sum_{t\in T} r_t$. Source conventions determined interval inclusion. For 19 systems, T contained only days with at least one recorded event. There were 5 systems that supplied regular snapshots aggregated into consecutive daily intervals, including zero-event days (metrla, pemsbay, montevideo, windmill, and wiki) and 1 system (chickenpox) that computed a daily-average rate at each weekly report date. Each S served as the realization of the system's dynamical process across its observation window.

Maximum Rate Target

The target value for each system was the log-transformed maximum rate realized across its observation window. The log-transformation compressed the range of maxima. For each rate series S on interval set T (Eq. (2)), the maximum rate $r_{\max}$ and log-transformed maximum rate target y^* are formally defined as:

$$y^* = \ln(1 + r_{\max}), \qquad r_{\max} = \max_{t \in T} r_t \tag{3}$$

Maxima can grow with observation window length [38–40]. Observation windows varied across sources and were held at their recorded durations. All frontiers remained conditional on their recorded observational windows (Supplementary Table 1). Each y^* served as the target bounded by the capacity frontier across its observation window.

Encoding and Standardization

Each network substrate was encoded using 21 graph invariants across 5 categories: simple, cohesion, extremal, statistical, and spectral. The mapping $G \to x$ is formally defined as:

$$x = \phi(G) = (x_1, x_2, \ldots, x_{21}) \in \mathbb{R}^{21} \tag{4}$$

For the 4 complete bipartite substrates (college, email, mooc, overflow), invariants were evaluated using exact analytical expressions for $K_{m,n}$ to avoid instantiating billions of edges.

Each dynamical process was encoded using 7 process signatures across 5 categories: dependence, dispersion, volatility, trend, and asymmetry. The mapping $S \to z$ is formally defined as:

$$z = \psi(S) = (z_1, z_2, \ldots, z_7) \in \mathbb{R}^7 \tag{5}$$

6 process signatures were scale-free. The Fano factor retained count scale, equaling the series mean $\bar{r}$ multiplied by the squared coefficient of variation, reflecting process dispersion.

x and z were centered by in-sample means and scaled by in-sample population standard deviations, with 0 standard deviations replaced by 1. The standardized vectors are formally defined as:

$$x' = \frac{x - \mu_x}{\sigma_x}, \qquad z' = \frac{z - \mu_z}{\sigma_z} \tag{6}$$

Means and scales were fit strictly in-sample (Supplementary Tables 7–8).

All features met the admission criteria: classical, deterministic, global to the network or series, free of arbitrary thresholds, without external parameters, and finite.

Frontier Learning

The capacity frontier was learned through a two-stage log-additive formulation. The structural stage (stage 1) was $f_C(x')$, and the process stage (stage 2) was $f_R(z')$. The same estimator type was trained separately in each stage. Hyperparameters were held constant. The estimated frontier value $\widehat{y}^*$ is formally defined as:

$$\widehat{y}^* = F(x', z') = f_C(x'; \tau_C) + f_R(z'; \tau_R) \tag{7}$$

where log-transformed maximum rates are bounded by this envelope ($y^* \leq \widehat{y}^*$), subject to slack and potential violations.

Stage 1 trained the structural component $f_C(x';\tau_C)$ to target y^* via asymmetric regression at $\tau_C = 0.99$. It placed the learned function near the upper-envelope given the graph invariants [37]. Stage 2 trained the process component $f_R(z';\tau_R)$ to the signed slack $y^* - f_C(x';\tau_C)$ at $\tau_R = 0.50$. Subtracting the process stage's training mean μ_R fixed its additive offset [41], resulting in the centered contribution $f_R(z';\tau_R)$ while leaving $f_C(x';\tau_C)$ unchanged. Summing both stages yielded the frontier estimate $\widehat{y}^*$. τ denotes the active stage's asymmetry level τ_C (stage 1) and τ_R (stage 2).

The linear parametric paradigm included linear quantile regression [34] minimizing mean pinball loss with ℓ_1-penalized slopes ($\alpha = 0.1$), unpenalized intercepts in both stages, and the `HiGHS 1.8.0` solver. Linear Asymmetric Weighted Least Squares (LAWS) minimized $(2n)^{-1}\sum_{i=1}^{n} w_i e_i^2 + \alpha\|\beta\|_1$ ($\alpha = 0.1$), with $w_i = \tau$ for $e_i \geq 0$ and $1-\tau$ otherwise [42]. Here e_i is the active stage's observed training response minus its prediction, n the training sample size, and β the slope vector. Cyclic coordinate descent used zero initial slopes, tolerance 10^{-4}, and at most 1,000 passes [43]. LAWS used an unpenalized stage 1 intercept initialized at the target mean and no stage 2 intercept. The concave-envelope regressor fitted a stage-specific linear quantile score ($\alpha = 0.1$) and interpolated the concave upper-envelope of target values with constant extrapolation beyond endpoints.

The non-linear ensemble paradigm included a quantile random forest with 100 fully grown bootstrap trees, squared-error splits, a minimum leaf size of 1, and all predictors eligible at each split [44]. Forest predictions were quantiles of the fitted conditional target distribution. Gradient-boosted quantile regression minimized pinball loss using 100 trees with maximum depth 3, learning rate 0.1, and the full training sample [45]. XGBoost minimized pinball loss with histogram tree construction, 100 trees, maximum depth 6, learning rate 0.3, an ℓ_2 penalty $\lambda = 1$, and full row and feature sampling [46]. Each estimator used the quantile assigned to the active stage.

The neural network paradigm included an MLP with pinball loss, an MLP with expectile loss, and an input-convex network with pinball loss [47]. Non-input weights in the convex network were made non-negative through absolute values. All networks used hidden layers of 8 and 4 units, ReLU activations [48], dropout of 0.1 [49], and output biases initialized at the active stage's training quantile. Full-batch AdamW optimization used learning rate 0.1, weight decay 0.01, gradient-norm clipping at 5.0, and 1,000 epochs [50]. Each stage's predictions were clipped to its own training target range before process centering and summation.

Performance Metrics

Frontier efficiency was measured using the Efficiency Index (EI). EI is the geometric mean of the non-violation rate $(1 - \mathrm{VR})$, the bounded mean-violation score $(1 + \mathrm{MV}/\overline{|y^*|})^{-1}$, and the bounded mean-slack score $(1 + \mathrm{MS}/\overline{|y^*|})^{-1}$, formally defined as:

$$\mathrm{EI} \;=\; \left[(1-\mathrm{VR})\cdot\left(1+\mathrm{MV}/\overline{|y^*|}\right)^{-1}\cdot\left(1+\mathrm{MS}/\overline{|y^*|}\right)^{-1}\right]^{1/3} \in [0,1] \tag{8}$$

For n evaluated systems with observed targets y_i^* and predicted values $\widehat{y}_i^*$, let $\mathcal{V} = \{i : y_i^* > \widehat{y}_i^*\}$ denote the violators. The Violation Rate (VR) is the fraction of evaluated systems exceeding their predictions, $\mathrm{VR} = |\mathcal{V}|/n$. The Mean Violation (MV) is the mean exceedance among violators, $\mathrm{MV} = \left(\sum_{i\in\mathcal{V}}(y_i^* - \widehat{y}_i^*)\right)/|\mathcal{V}|$ (0 when $|\mathcal{V}| = 0$). The Mean Slack (MS) is the mean non-negative gap between $\widehat{y}_i^*$ and y_i^* across all evaluated systems, $\mathrm{MS} = n^{-1}\sum_{i=1}^{n}\max(\widehat{y}_i^* - y_i^*, 0)$. MV and MS are normalized by the sample mean absolute target $\overline{|y^*|} = n^{-1}\sum_{i=1}^{n}|y_i^*|$. For numerical stability, 10^{-10} was added to the normalization denominator and used as a floor for $1 - \mathrm{VR}$. MV and MS are expressed in natural-log units. Note that $\mathrm{EI} = 1$ under perfect frontier efficiency, which requires no violations and no slack ($\mathrm{VR} = \mathrm{MV} = \mathrm{MS} = 0$).

Frontier consensus was measured using the Consensus Index (CI). CI is the geometric mean of rescaled Spearman rank correlation $\rho_s^+ = (1 + \rho_s)/2$ [51], top-weighted Rank-Biased Overlap (RBO) [52], and distance correlation (dCor) [53], formally defined as:

$$\mathrm{CI} \;=\; \left[\rho_s^+ \cdot \mathrm{RBO} \cdot \mathrm{dCor}\right]^{1/3} \in [0, 1] \tag{9}$$

ρ_s used average ranks for ties and was set to 0 when either vector was constant, yielding $\rho_s^+ = 0.5$. RBO evaluated descending rankings with persistence $p_{\mathrm{RBO}} = 0.9$. Ties were resolved by ascending input index. RBO used tail extrapolation from agreement at the last evaluated depth. dCor used the biased V-statistic sample estimator with Euclidean-distance exponent 1 and 0 for a constant vector. All CI components were clipped $[0, 1]$ against floating point imprecision. CI measured estimation-observation consensus between predicted values $\widehat{y}^*$ and observed targets y^* as well as pairwise model consensus between predicted values $\widehat{y}_i^*$ and $\widehat{y}_j^*$ from distinct estimators $i \neq j$. Note that CI = 1 under perfect frontier consensus, which requires that all three components be in perfect agreement ($\rho_s^+ = \mathrm{RBO} = \mathrm{dCor} = 1$).

EI and CI serve as paired metrics designed to measure complementary aspects of frontier performance. EI measures boundary containment and envelope tightness by penalizing violations and excessive slack. CI measures system-specific fidelity by measuring monotonic ordering, top-capacity overlap, and statistical dependence. Critically, neither metric is sufficient in isolation. An arbitrarily high and uninformative constant frontier above all observations can achieve a noteworthy EI despite its slack, yet degrades consensus. Conversely, an uncalibrated ranking can achieve high CI while failing to bound the data. A valid capacity frontier must simultaneously demonstrate containment and fidelity.

Resampling and Leakage Control

Generalization was evaluated using 4 resampling regimes: repeated 5-Fold-CV, repeated 10-Fold-CV, domain-level LOGO-CV, and discipline-level LOGO-CV. Repeated k-Fold-CV used random reshuffling over 30 repetitions [54]. Domain-level LOGO-CV withheld 1 of the 5 broad domains for each domain. Discipline-level LOGO-CV withheld 1 of the 25 disciplines for each discipline [55–59]. LOGO-CV folds were deterministic. Repetitions were drawn across training seeds $42 + b$ for $b = 0, \ldots, 29$.

Within each fold, standardizers for x and z (Eq. (6)), both estimators $f_C(.)$ and $f_R(.)$, and the process centering mean μ_R (Eq. (7)) were fit *exclusively* on the training set. Estimators were freshly cloned on every fold. At inference time, models $f_C(x')$ and $f_R(z')$ had no access to the target y^* or raw rate series S. Recall that y^* was log-transformed, which did not require parameter fitting. Therefore, information from out-of-sample systems did not enter standardization and model training on the original corpus during cross-validation.

Falsified negative-control corpora were generated from the full corpus, and feature-wise sample standard deviations for additive-noise perturbations were computed across the full corpus before cross-validation. These corpus-wide operations specified the negative controls and scales related to additive-noise. Across all cross-validated stress tests, standardizers and estimators were fit *exclusively* on the corresponding training set.

Perturbation, Falsification, Ablation

Network substrate perturbations at intensity ϵ modified $G = (V, E)$ (Eq. (1)) across prespecified parameter grids and protocols (Supplementary Table 9). Rewiring used $\widetilde{G} = \mathcal{R}_q(G)$, where $\mathcal{R}_q$ performed $q = \max(1, \lfloor \epsilon |E| \rfloor)$ degree-preserving trials on a simple graph [60]. Densification used

$\widetilde{G} = (V, E \cup A)$, where A was sampled uniformly without replacement from $\binom{V}{2} \setminus E$, with $|A| = \min\{\lfloor \epsilon|E| \rfloor, \binom{|V|}{2} - |E|\}$ [61]. Node removal used the induced subgraph $\widetilde{G} = G[V \setminus U]$, where U was sampled uniformly without replacement from V and $|U| = \lfloor \epsilon|V| \rfloor$ [62, 63]. Graph invariants were recomputed as $\widetilde{x} = \phi(\widetilde{G})$ (Eq. (4)). For the 4 complete bipartite substrates, node removal used exact closed-form invariants. Rewiring and densification used deterministic approximations (Supplementary Note 1).

Dynamical process perturbations modified the rate series through power-law scaling, trailing-mean smoothing, and bootstrapping. For these expressions, r_ℓ denotes the rate at chronological position $\ell = 1, \ldots, L$, where $L = |T|$. Scaling used $\widetilde{r}_\ell = (r_\ell + 1)^\gamma - 1$. Smoothing used $\widetilde{r}_\ell = (\sum_{h=\max(1,\ell-w+1)}^{\ell} r_h)/\min(w, \ell)$, shortening the w-interval window at the beginning of the series. Both methods were deterministic. Bootstrapping drew independent indices $I_h \sim \mathrm{Unif}\{1, \ldots, L\}$ for $h = 1, \ldots, k$ [64], with $k = \max(1, \mathrm{round}(sL))$ and halves rounded to even. The first k intervals were $\widetilde{r}_\ell = r_{I_\ell}$ for $\ell \leq k$. The remaining intervals were $\widetilde{r}_\ell = r_{I_{J_\ell}}$ for $\ell > k$, with independent $J_\ell \sim \mathrm{Unif}\{1, \ldots, k\}$. Process signatures were recomputed as $\widetilde{z} = \psi(\widetilde{S})$ (Eq. (5)), where $\widetilde{S}$ was the modified series. All perturbations retained the original targets.

Graph invariants and process signatures were also perturbed through additive noise, multiplicative jitter, and coordinate subsetting. These methods were applied separately to x and z prior to standardization. For either vector $v \in \{x, z\}$, additive noise used $\widetilde{v}_j = v_j + \varepsilon_j$, with $\varepsilon_j \sim \mathcal{N}(0, (s\sigma_j)^2)$ and σ_j the corpus-wide sample standard deviation of coordinate j. Jitter used $\widetilde{v}_j = \max\{v_j(1+\eta_j), 0\}$ for originally non-negative v_j and $\widetilde{v}_j = v_j(1 + \eta_j)$ otherwise, with $\eta_j \sim \mathcal{N}(0, s^2)$. Masking used $\widetilde{v}_j = v_j \mathbf{1}[j \in M]$, where M was a uniformly sampled subset of retained coordinates with $|M| = \lfloor \rho p \rfloor$. Here ρ was the retention fraction, $p = 21$ for x or $p = 7$ for z. Each system used the same seed within a realization, sharing coordinate-wise noise and jitter draws and the masked subset across systems.

The 3 falsification procedures deliberately destroyed the mapping $(x', z') \to y^*$. Let $i = 1, \ldots, 25$ index systems and D_i denote their broad domains. Target permutation used $\widetilde{y}_i^* = y^*_{\pi(i)}$, where π was a permutation satisfying $D_{\pi(i)} \neq D_i$, preserving each system's original x and z and the target's marginal distribution. For either raw vector $v \in \{x, z\}$, random generation independently sampled $\widetilde{v}_{ij} \sim \mathcal{U}(a_j, b_j)$, where $a_j = \min_h v_{hj}$ and $b_j = \max_h v_{hj}$ across the corpus. Vector generation resampled empirical coordinates with replacement [64] and added Gaussian noise as $\widetilde{v}_{ij} = v_{B_{ij},j} + \xi_{ij}$, with independent $B_{ij} \sim \mathrm{Unif}\{1, \ldots, 25\}$ and $\xi_{ij} \sim \mathcal{N}(0, \nu_j^2)$. Here ν_j was the coordinate's corpus-wide population standard deviation, floored at 10^{-10}. Both feature-generation procedures retained each system's original target.

The 4 ablations compared alternative specifications with the log-additive decomposition using the original training folds and standard estimator protocols. Invariants only used $F_{\mathrm{Inv}}(x') = f_{\mathrm{Inv}}(x'; \tau_C)$. Signatures only used $F_{\mathrm{Sig}}(z') = f_{\mathrm{Sig}}(z'; \tau_C)$. Joint features used $F_{\mathrm{Joint}}(x', z') = f_{\mathrm{Joint}}([x', z']; \tau_C)$, where brackets denote vector concatenation. These ablations used single-stage fits at $\tau_C = 0.99$. Interaction terms formed a 147-coordinate vector $h = (x_j z_k)_{j=1,\ldots,21\,;\,k=1,\ldots,7}$ from raw coordinates and standardized it within each training fold to h'. The alternative frontier was $F_{\mathrm{Int}}(x', z', h') = f_{C,\mathrm{Int}}(x'; \tau_C) + f_{R,\mathrm{Int}}([z', h']; \tau_R)$, with both stages refit and $f_{R,\mathrm{Int}}$ centered (Eq. (7)).

Evaluation and Statistical Inference

Inference used 45 estimator-by-domain comparisons summarized over 30 repetitions. Within each domain, EI was computed using seed-averaged predictions for resampling equivalence and falsification. Perturbation and ablation used repetition means in both the original and tested conditions. CI used seed-averaged predictions except for stochastic perturbations, where metrics were averaged after computing each paired training seed and realization.

Paired differences ΔEI and ΔCI were computed as test comparison minus baseline and summarized by their median. Baselines were domain-level LOGO-CV for resampling equivalence, the original condition for perturbation and falsification, and the log-additive decomposition for ablation. All comparisons used One-Sided Wilcoxon Signed-Rank tests [65], assuming symmetric paired-difference distributions with location parameter θ. Inference was conditional on the given panel whose shared estimators and cross-validation folds induced dependence that may affect p-value calibration.

For a margin $\delta > 0$, resampling equivalence and perturbation stability used TOST [66] of $H_{01} : \theta \leq -\delta$ and $H_{02} : \theta \geq \delta$. Equivalence required rejection of both nulls with the TOST p-value taken as the larger of the 2 one-sided values. Falsification tested degradation ($H_0 : \theta \geq 0$ versus $H_1 : \theta < 0$). Ablation tested non-inferiority of the log-additive decomposition to each alternative [67] ($H_0 : \theta \geq \delta$ versus $H_1 : \theta < \delta$).

All margins δ were empirically derived based on interquartile ranges (IQR) rounded down 2 decimal places (Supplementary Tables 3–4 and 6). Holm-Bonferroni adjustments (Holm-adjusted p, denoted p_{H}) [68] used a significance threshold of $\alpha = 0.05$ separately for EI and CI within 5 families: resampling, perturbation, falsification, simple ablations, and complex ablations.

Rank-biserial effect sizes were computed as $r = (W_+ - W_-)/(W_+ + W_-)$ from nonzero, unshifted paired differences [69]. Here W_+ and W_- are the sums of ranks of absolute differences with positive and negative signs, respectively, with average ranks for ties.

Full-corpus consensus tests were descriptive (Fig. 3). Each estimator's predictions were averaged across training seeds. For stochastic perturbations, realizations were paired before computing estimation-observation and pairwise model CI. Pairwise model consensus was the median CI across the 36 distinct estimator pairs. Original data fits were frozen for evaluation on the highest intensity perturbed corpora and the falsified corpora. Ablations were refit on the full original corpus.

Implementation and Reproducibility

All analyses were implemented in `Python 3.11.9` [70, 71]. All dependency versions were pinned in the repository's `requirements.txt`. Graphs, processes, and encodings used `python-igraph 1.0.0` [72, 73], `NumPy 2.3.5` [74, 75], `SciPy 1.16.3` [76, 77], and `pandas 3.0.0` [78, 79]. Urban street network construction used `networkx 3.5` [80, 81] and `OSMnx 2.0.6` [82, 83].

Public dataset retrieval was conducted using `requests 2.32.5` [84]. Temporal graph datasets were loaded using `torch-geometric 2.7.0` [85, 86] and `torch_geometric_temporal 0.56.2` [87, 88]. Observational records for gravitational wave and meteorological systems were retrieved using `gwosc 0.8.1` and `meteostat 1.7.6`, respectively [89, 90].

Estimators used `scikit-learn 1.8.0` [91, 92], `sklearn-quantile 0.1.1` [93], `XGBoost 3.1.1` [46, 94], and `PyTorch 2.9.0` [95, 96]. Distance correlation was computed using `dcor 0.7` [97], and scientific figures were generated with `matplotlib 3.9.2` [98, 99]. Unspecified settings followed library defaults.

All stochastic procedures started from random seed 42 with repeat indices applied as fixed numerical offsets. Random permutations, cross-validation folds, and perturbation realizations originated from seeded generators.

`PyTorch 2.9.0` deterministic algorithms were executed in warn-only mode and execution defaulted to CUDA when GPU resources were available and CPU otherwise. Results reproduced with the code repository, corpus, and environment specifications are subject to floating-point imprecision.

Data Availability

The corpus was collated from publicly available data sources (Supplementary Table 1). The processed records used in all analyses are distributed with the source code repository at: https://github.com/nickkunz/network-capacity.

Code Availability

The scripts for data processing, frontier estimation, evaluation, perturbation, falsification, and ablation, along with the notebooks that reproduce every figure and table are also distributed with the source code repository at: https://github.com/nickkunz/network-capacity.

Acknowledgements

We would like to thank F. Parise, W. Xu, N. de Santi, and S. Visvesvaran for their early feedback, discussions, and encouragement.

Author Contributions

N. Kunz initiated and developed the concept, methods, implementation, testing, documentation, and manuscript. H. O. Gao initiated and supervised the development of the investigation and manuscript. The authors have agreed to the submitted version of this manuscript.

Funding

This work was supported by Cornell University and Gao Labs.

Competing Interests

The authors declare no competing interests.

Additional Information

Supplementary information is in Supplementary Tables 1–9 and Supplementary Notes 1–2.

Correspondence and requests for materials should be addressed to N. Kunz.

References

[1] M. E. J. Newman. The structure and function of complex networks. *SIAM Review*, 45:167–256, 2003.

[2] Marten Scheffer et al. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009.

[3] Jian Gao, Baruch Barzel, and Albert-László Barabási. Universal resilience patterns in complex networks. *Nature*, 530:307–312, 2016.

[4] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: an explanation of the 1/f noise. *Physical Review Letters*, 59:381–384, 1987. doi: 10.1103/PhysRevLett.59.381.

[5] Sergey V. Buldyrev, Roni Parshani, Gerald Paul, H. Eugene Stanley, and Shlomo Havlin. Catastrophic cascade of failures in interdependent networks. *Nature*, 464:1025–1028, 2010. doi: 10.1038/nature08932.

[6] Sui Huang, G. S. Eichler, Yaneer Bar-Yam, and Donald E. Ingber. Cell fates as high-dimensional attractor states of a complex gene regulatory network. *Physical Review Letters*, 94:128701, 2005. doi: 10.1103/PhysRevLett.94.128701.

[7] Roger Guimerà, Albert Díaz-Guilera, Fernando Vega-Redondo, Antonio Cabrales, and Alex Arenas. Optimal network topologies for local search with congestion. *Physical Review Letters*, 89:248701, 2002.

[8] Daqing Li et al. Percolation transition in dynamical traffic network with evolving critical bottlenecks. *Proceedings of the National Academy of Sciences*, 112:669–672, 2015.

[9] L. R. Taylor. Aggregation, variance and the mean. *Nature*, 189:732–735, 1961.

[10] Zoltán Eisler, Imre Bartos, and János Kertész. Fluctuation scaling in complex systems: Taylor's law and beyond. *Advances in Physics*, 57:89–142, 2008. doi: 10.1080/00018730801893043.

[11] Baruch Barzel and Albert-László Barabási. Universality in network dynamics. *Nature Physics*, 9:673–681, 2013.

[12] Stefano Boccaletti, Vito Latora, Yamir Moreno, Mario Chavez, and Dong-U Hwang. Complex networks: structure and dynamics. *Physics Reports*, 424:175–308, 2006.

[13] Lester R. Ford, Jr. and Delbert R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956. doi: 10.4153/CJM-1956-045-5.

[14] Rudolf Ahlswede, Ning Cai, Shuo-Yen Robert Li, and Raymond W. Yeung. Network information flow. *IEEE Transactions on Information Theory*, 46:1204–1216, 2000.

[15] Piyush Gupta and P. R. Kumar. The capacity of wireless networks. *IEEE Transactions on Information Theory*, 46:388–404, 2000. doi: 10.1109/18.825799.

[16] Z. Y. Chen and X. F. Wang. Effects of network structure and routing strategy on network capacity. *Physical Review E*, 73:036107, 2006. doi: 10.1103/PhysRevE.73.036107.

[17] Romualdo Pastor-Satorras and Alessandro Vespignani. Epidemic spreading in scale-free networks. *Physical Review Letters*, 86:3200–3203, 2001.

[18] Adilson E. Motter and Ying-Cheng Lai. Cascade-based attacks on complex networks. *Physical Review E*, 66:065102, 2002.

[19] Romualdo Pastor-Satorras, Claudio Castellano, Piet Van Mieghem, and Alessandro Vespignani. Epidemic processes in complex networks. *Reviews of Modern Physics*, 87:925–979, 2015. doi: 10.1103/RevModPhys.87.925.

[20] Alex Arenas, Albert Díaz-Guilera, Jürgen Kurths, Yamir Moreno, and Changsong Zhou. Synchronization in complex networks. *Physics Reports*, 469:93–153, 2008.

[21] Yang-Yu Liu, Jean-Jacques Slotine, and Albert-László Barabási. Controllability of complex networks. *Nature*, 473:167–173, 2011. doi: 10.1038/nature10011.

[22] Marc Barthélemy. Spatial networks. *Physics Reports*, 499:1–101, 2011. doi: 10.1016/j.physrep.2010.11.002.

[23] Alain Barrat, Marc Barthélemy, and Alessandro Vespignani. *Dynamical Processes on Complex Networks*. Cambridge University Press, 2008.

[24] Carter T. Butts. Revisiting the foundations of network analysis. *Science*, 325:414–416, 2009. doi: 10.1126/science.1171022.

[25] Petter Holme and Jari Saramäki. Temporal networks. *Physics Reports*, 519:97–125, 2012.

[26] Geoffrey B. West, James H. Brown, and Brian J. Enquist. A general model for the origin of allometric scaling laws in biology. *Science*, 276:122–126, 1997.

[27] Luís M. A. Bettencourt, José Lobo, Dirk Helbing, Christian Kühnert, and Geoffrey B. West. Growth, innovation, scaling, and the pace of life in cities. *Proceedings of the National Academy of Sciences*, 104:7301–7306, 2007.

[28] Dirk Brockmann, Lars Hufnagel, and Theo Geisel. The scaling laws of human travel. *Nature*, 439:462–465, 2006.

[29] M. J. Farrell. The measurement of productive efficiency. *Journal of the Royal Statistical Society: Series A*, 120:253–290, 1957. doi: 10.2307/2343100.

[30] Léopold Simar and Paul W. Wilson. Sensitivity analysis of efficiency scores: how to bootstrap in nonparametric frontier models. *Management Science*, 44:49–61, 1998. doi: 10.1287/mnsc.44.1.49.

[31] Dennis Aigner, C. A. Knox Lovell, and Peter Schmidt. Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics*, 6:21–37, 1977.

[32] Wim Meeusen and Julien van den Broeck. Efficiency estimation from cobb-douglas production functions with composed error. *International Economic Review*, 18:435–444, 1977.

[33] Abraham Charnes, William W. Cooper, and Edwardo Rhodes. Measuring the efficiency of decision making units. *European Journal of Operational Research*, 2:429–444, 1978.

[34] Roger Koenker and Gilbert Bassett, Jr. Regression quantiles. *Econometrica*, 46:33–50, 1978.

[35] Subal C. Kumbhakar and C. A. Knox Lovell. *Stochastic Frontier Analysis*. Cambridge University Press, 2000.

[36] Catherine Cazals, Jean-Pierre Florens, and Léopold Simar. Nonparametric frontier estimation: a robust approach. *Journal of Econometrics*, 106:1–25, 2002. doi: 10.1016/S0304-4076(01)00080-X.

[37] Victor Chernozhukov. Extremal quantile regression. *The Annals of Statistics*, 33:806–839, 2005. doi: 10.1214/009053604000001165.

[38] Emil J. Gumbel. *Statistics of Extremes*. Columbia University Press, 1958.

[39] James Pickands, III. Statistical inference using extreme order statistics. *The Annals of Statistics*, 3:119–131, 1975. doi: 10.1214/aos/1176343003.

[40] Abdelaati Daouia, Jean-Pierre Florens, and Léopold Simar. Frontier estimation and extreme value theory. *Bernoulli*, 16:1039–1063, 2010. doi: 10.3150/10-BEJ256.

[41] Trevor Hastie and Robert Tibshirani. Generalized additive models. *Statistical Science*, 1: 297–310, 1986.

[42] Whitney K. Newey and James L. Powell. Asymmetric least squares estimation and testing. *Econometrica*, 55:819–847, 1987.

[43] Jerome Friedman, Trevor Hastie, and Robert Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33:1–22, 2010.

[44] Nicolai Meinshausen. Quantile regression forests. *Journal of Machine Learning Research*, 7: 983–999, 2006.

[45] Jerome H. Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29:1189–1232, 2001.

[46] Tianqi Chen and Carlos Guestrin. Xgboost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[47] Brandon Amos, Lei Xu, and J. Zico Kolter. Input convex neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 146–155, 2017.

[48] Vinod Nair and Geoffrey E. Hinton. Rectified linear units improve restricted boltzmann machines. In *Proceedings of the 27th International Conference on Machine Learning*, pages 807–814, 2010.

[49] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15:1929–1958, 2014.

[50] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

[51] Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15:72–101, 1904.

[52] William Webber, Alistair Moffat, and Justin Zobel. A similarity measure for indefinite rankings. *ACM Transactions on Information Systems*, 28:20, 2010.

[53] Gábor J. Székely, Maria L. Rizzo, and Nail K. Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, 35:2769–2794, 2007.

[54] Prabir Burman. A comparative study of ordinary cross-validation, v-fold cross-validation and the repeated learning-testing methods. *Biometrika*, 76:503–514, 1989.

[55] Krikamol Muandet, David Balduzzi, and Bernhard Schölkopf. Domain generalization via invariant feature representation. In *Proceedings of the 30th International Conference on Machine Learning*, pages 10–18, 2013.

[56] Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society: Series B*, 78:947–1012, 2016. doi: 10.1111/rssb.12167.

[57] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk minimization, 2019.

[58] David R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40:913–929, 2017.

[59] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2021. doi: 10.48550/arXiv.2007.01434.

[60] Sergei Maslov and Kim Sneppen. Specificity and stability in topology of protein networks. *Science*, 296:910–913, 2002.

[61] Paul Erdős and Alfréd Rényi. On the evolution of random graphs. *Publications of the Mathematical Institute of the Hungarian Academy of Sciences*, 5:17–61, 1960.

[62] Réka Albert, Hawoong Jeong, and Albert-László Barabási. Error and attack tolerance of complex networks. *Nature*, 406:378–382, 2000. doi: 10.1038/35019019.

[63] Reuven Cohen, Keren Erez, Daniel ben Avraham, and Shlomo Havlin. Resilience of the internet to random breakdowns. *Physical Review Letters*, 85:4626–4628, 2000. doi: 10.1103/PhysRevLett.85.4626.

[64] Bradley Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7: 1–26, 1979.

[65] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1:80–83, 1945.

[66] Donald J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15:657–680, 1987.

[67] Stefan Wellek. *Testing Statistical Hypotheses of Equivalence and Noninferiority*. CRC Press, 2 edition, 2010.

[68] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6:65–70, 1979.

[69] Dave S. Kerby. The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology*, 3:11.IT.3.1, 2014. doi: 10.2466/11.IT.3.1.

[70] Guido van Rossum and Fred L. Drake. *Python 3 Reference Manual*. CreateSpace, 2009.

[71] Python. Cpython, version 3.11.9, 2024. URL https://github.com/python/cpython/releases/tag/v3.11.9.

[72] Gábor Csárdi and Tamás Nepusz. The igraph software package for complex network research. *InterJournal Complex Systems*, page 1695, 2006.

[73] igraph. igraph: the network analysis package, python-igraph 1.0.0, 2025. URL https://github.com/igraph/python-igraph/releases/tag/1.0.0.

[74] Charles R. Harris et al. Array programming with numpy. *Nature*, 585:357–362, 2020.

[75] NumPy. Numpy, version 2.3.5, 2025. URL https://github.com/numpy/numpy/releases/tag/v2.3.5.

[76] Pauli Virtanen et al. Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods*, 17:261–272, 2020.

[77] SciPy. Scipy, version 1.16.3, 2025. URL https://github.com/scipy/scipy/releases/tag/v1.16.3.

[78] Wes McKinney. Data structures for statistical computing in python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010.

[79] pandas. pandas, version 3.0.0, 2026.

[80] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. Exploring network structure, dynamics, and function using networkx. In *Proceedings of the 7th Python in Science Conference*, pages 11–15, 2008. doi: 10.25080/TCWV9851.

[81] NetworkX. Networkx, version 3.5, 2025. URL https://github.com/networkx/networkx/releases/tag/networkx-3.5.

[82] Geoff Boeing. Osmnx, version 2.0.6, 2025. URL https://pypi.org/project/osmnx/2.0.6/.

[83] Geoff Boeing. Modeling and analyzing urban networks and amenities with OSMnx. *Geographical Analysis*, 57:567–577, 2025. doi: 10.1111/gean.70009.

[84] Kenneth Reitz. Requests, version 2.32.5, 2025. URL https://github.com/psf/requests/releases/tag/v2.32.5.

[85] Matthias Fey and Jan E. Lenssen. Fast graph representation learning with pytorch geometric, 2019.

[86] PyTorch Geometric. Pytorch geometric, version 2.7.0, 2025. URL https://github.com/pyg-team/pytorch_geometric/releases/tag/2.7.0.

[87] Benedek Rozemberczki et al. Pytorch geometric temporal: spatiotemporal signal processing with neural machine learning models. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management*, pages 4564–4573, 2021.

[88] PyTorch Geometric Temporal. Pytorch geometric temporal, version 0.56.2, 2025. URL https://pypi.org/project/torch-geometric-temporal/0.56.2/.

[89] Duncan Macleod and Martin Beroiz. gwosc: A python interface to the GW open science data archive, version 0.8.1, 2025. URL https://pypi.org/project/gwosc/0.8.1/.

[90] Meteostat. Meteostat python package, version 1.7.6, 2025. URL https://github.com/meteostat/meteostat/releases/tag/v1.7.6.

[91] Fabian Pedregosa et al. Scikit-learn: machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[92] scikit-learn. scikit-learn, version 1.8.0, 2025.

[93] Jasper Roebroek. sklearn-quantile: quantile regression models for scikit-learn, version 0.1.1, 2025. URL https://github.com/jasperroebroek/sklearn-quantile.

[94] XGBoost. Xgboost: scalable gradient boosting, version 3.1.1, 2025. URL https://github.com/dmlc/xgboost/releases/tag/v3.1.1.

[95] Adam Paszke et al. Pytorch: an imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, pages 8024–8035, 2019.

[96] PyTorch. Pytorch, version 2.9.0, 2025. URL https://github.com/pytorch/pytorch/releases/tag/v2.9.0.

[97] Carlos Ramos Carreño. dcor: distance correlation and dependence measures in python, version 0.7, 2026. URL https://pypi.org/project/dcor/0.7/.

[98] John D. Hunter. Matplotlib: a 2d graphics environment. *Computing in Science and Engineering*, 9:90–95, 2007.

[99] Matplotlib. Matplotlib, version 3.9.2, 2024. URL https://github.com/matplotlib/matplotlib/releases/tag/v3.9.2.

[100] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of small-world networks. *Nature*, 393:440–442, 1998.

[101] Mark E. J. Newman. Assortative mixing in networks. *Physical Review Letters*, 89:208701, 2002.

[102] Stephen B. Seidman. Network structure and minimum degree. *Social Networks*, 5:269–287, 1983.

[103] David W. Matula and Leland L. Beck. Smallest-last ordering and clustering and graph coloring algorithms. *Journal of the ACM*, 30:417–427, 1983.

[104] Fan R. K. Chung. *Spectral Graph Theory*. American Mathematical Society, 1997.

[105] Ernesto Estrada and Desmond J. Higham. Network properties revealed through matrix functions. *SIAM Review*, 52:696–714, 2010.

[106] Matthias Dehmer and Abbe Mowshowitz. A history of graph entropy measures. *Information Sciences*, 181:57–78, 2011. doi: 10.1016/j.ins.2010.08.041.

[107] Sergey N. Dorogovtsev and José F. F. Mendes. Evolution of networks. *Advances in Physics*, 51: 1079–1187, 2002.

[108] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.

[109] Jae Dong Noh and Heiko Rieger. Random walks on complex networks. *Physical Review Letters*, 92:118701, 2004. doi: 10.1103/PhysRevLett.92.118701.

[110] Piet Van Mieghem. *Graph Spectra for Complex Networks*. Cambridge University Press, 2011.

[111] Harold E. Hurst. Long-term storage capacity of reservoirs. *Transactions of the American Society of Civil Engineers*, 116:770–799, 1951.

[112] Benoit B. Mandelbrot and John R. Wallis. Robustness of the rescaled range r/s in the measurement of noncyclic long run statistical dependence. *Water Resources Research*, 5:967–988, 1969. doi: 10.1029/WR005i005p00967.

[113] Ugo Fano. Ionization yield of radiations. ii. the fluctuations of the number of ions. *Physical Review*, 72:26–29, 1947.

[114] Ben D. Fulcher and Nicholas S. Jones. Highly comparative feature-based time-series classification. *IEEE Transactions on Knowledge and Data Engineering*, 26:3026–3037, 2014.

[115] Carl H. Lubba et al. catch22: Canonical time-series characteristics. *Data Mining and Knowledge Discovery*, 33:1821–1852, 2019.

[116] C.-K. Peng et al. Mosaic organization of dna nucleotides. *Physical Review E*, 49:1685–1689, 1994. doi: 10.1103/PhysRevE.49.1685.

[117] Kun Hu, Plamen Ch. Ivanov, Zhi Chen, P. Carpena, and H. Eugene Stanley. Effect of trends on detrended fluctuation analysis. *Physical Review E*, 64:011114, 2001. doi: 10.1103/PhysRevE.64.011114.

[118] Kwang-Il Goh and Albert-László Barabási. Burstiness and memory in complex systems. *Europhysics Letters*, 81:48002, 2008. doi: 10.1209/0295-5075/81/48002.

[119] Albert-László Barabási. The origin of bursts and heavy tails in human dynamics. *Nature*, 435:207–211, 2005. doi: 10.1038/nature03459.

[120] Michael Molloy and Bruce Reed. A critical point for random graphs with a given degree sequence. *Random Structures and Algorithms*, 6:161–180, 1995.

Supplementary Tables

Supplementary Table 1. Corpus composition, definitions, sources, and date provenance.

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
gwosc	Earth & Physical Sciences	Cosmology	Gravitational-wave detectors (H1, L1, V1, K1, G1).	Complete graph among detectors.	GWTC event records from any GWTC catalog attributed to at least one corpus detector, including the marginal and auxiliary catalogs.	Maximum daily records $r_{\max} = 3$, $y^* = 1.386$.	2015-09-14 to 2024-01-09 ($n = 3{,}040$).	GWOSC event API (S1, S24, S25, S26, S27, S30).	Catalog articles published in 2019, 2021, 2023, and 2024. GWTC-4.0 first released 26 August 2025 and its results article was published in 2026. Fixed event cutoff 2024-01-09.
river	Earth & Physical Sciences	Hydrology	USGS stream gauges on the Colorado River (HUC 15010001, 15010002, 15010005).	Non-crossing path among gauges.	Peaks of discharge (parameter 00060) above the site-wise 99th percentile.	Maximum daily discharge-peak exceedances $r_{\max} = 9$, $y^* = 2.303$.	2014-11-12 to 2023-08-01 ($n = 3{,}185$).	USGS NWIS site and instantaneous-values services (S2).	NWIS dataset record published 1994. API query cutoff 2024-12-31.
auger	Earth & Physical Sciences	Physics	Pierre Auger Observatory surface-detector stations.	Reciprocal 6-nearest neighbours within 1.1× array spacing (1,500 m and 750 m arrays).	Surface-detector air-shower events.	Maximum daily air-shower events $r_{\max} = 66$, $y^* = 4.205$.	2004-01-03 to 2018-09-01 ($n = 5{,}356$).	Auger Open Data station map and Zenodo record 10488964 (S3).	Open Data version 3 released 2024. Observations span 2004–2018.
seismic	Earth & Physical Sciences	Geology	IRIS IU network stations.	Complete graph among stations.	Reviewed USGS events, 2023.	Maximum daily reviewed events $r_{\max} = 1{,}129$, $y^* = 7.030$.	2023-01-01 to 2023-12-30 ($n = 364$).	IRIS FDSN station service (S4) and USGS FDSN event service (S5).	GSN dataset record published 1988. ComCat record published 2017. Event-query cutoff 2023-12-31.
rain	Earth & Physical Sciences	Meteorology	Meteostat hourly stations in Laos.	Delaunay triangulation of station coordinates.	Station-hours with precipitation > 1 mm.	Maximum daily wet station-hours $r_{\max} = 46$, $y^* = 3.850$.	2024-01-12 to 2024-03-30 ($n = 79$).	Meteostat (S6).	Mutable service accessed 1 September 2026. Query cutoff 2024-03-31.
celegans	Life Sciences & Medicine	Biology	*C. elegans* developmental stages.	Known stage-to-stage transitions (normal, dauer, and recovery pathways), entered undirected.	Developmental transitions at published mean times.	Maximum daily developmental transitions $r_{\max} = 8$, $y^* = 2.197$.	Days 0–33 ($n = 34$).	Cassada & Russell, *Dev. Biol.* **46**, 326–342 (1975) (S7), and Golden & Riddle, *Dev. Biol.* **102**, 368–378 (1984) (S8).	Timing studies published 1975 and 1984. Days are developmental-age indices, not calendar dates.

Continued on next page

Supplementary Table 1. Corpus composition and provenance *(continued)*.

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
crop	Life Sciences & Medicine	Ecology	Crops and pollinator taxa.	Observed crop–pollinator interactions.	Daily flower-visitor census visits from primary study records.	Maximum daily census visits $r_{\max} = 1{,}266$, $y^* = 7.144$.	2004-05-07 to 2012-05-29 ($n = 2{,}945$).	CropPol repository (Rader et al. primary census records) (S9, S31).	Database article published 2022. Observations span 2004–2012.
chickenpox	Life Sciences & Medicine	Epidemiology	Hungarian counties.	County adjacency from the Chickenpox dataset.	Weekly county case counts summed across counties, divided by 7 to a daily average.	Maximum daily-average cases $r_{\max} = 378$, $y^* = 5.938$.	2005-01-03 to 2014-12-29 ($n = 3{,}648$).	Rozemberczki et al. Chickenpox dataset and UCI Hungarian chickenpox cases (S10).	UCI dataset released 2021. Observations span 2005–2014. Source reports weekly totals, converted to daily averages.
faers	Life Sciences & Medicine	Pharmacology	Medicinal products and MedDRA reaction terms co-reported with imatinib.	Drug–reaction co-occurrence.	Unique FAERS case receipts.	Maximum daily case receipts $r_{\max} = 45$, $y^* = 3.829$.	2004-01-23 to 2025-12-31 ($n = 8{,}014$).	openFDA drug-event API, query IMATINIB (S11).	Mutable API retrieved 13 March 2026. Case-receipt cutoff 2025-12-31.
epilepsy	Life Sciences & Medicine	Neurology	19 scalp electrodes (10–20 system).	Anatomical neighbours on the 10–20 montage.	Annotated seizures in CHB-MIT (chb01–chb24).	Maximum daily annotated seizures $r_{\max} = 27$, $y^* = 3.332$.	Relative recording days 0–76 ($n = 77$).	PhysioNet CHB-MIT (S12).	PhysioNet version 1.0.0 released 2010. Calendar dates are unavailable, so recording order is expressed as relative days.
federal	Trade & Institutions	Government	Awarding agencies and recipients.	Agency–recipient contract pairs.	Waterfowl-related contract start dates.	Maximum daily contract starts $r_{\max} = 17$, $y^* = 2.890$.	2011-01-19 to 2024-12-30 ($n = 5{,}095$).	USAspending API, keyword waterfowl (S13).	Mutable API retrieved 25 August 2026. Award-start cutoff 2024-12-31.
mooc	Trade & Institutions	Education	Students and course actions.	Complete bipartite graph between students and actions.	Student–action timestamps.	Maximum daily student actions $r_{\max} = 18{,}213$, $y^* = 9.810$.	Days 0–29 ($n = 30$).	SNAP act-mooc (S14).	Dataset article published 2019. Timestamps reindexed to relative days.
amazon	Trade & Institutions	Commerce	Reviewers and book products.	Reviewer–product review pairs.	Review timestamps.	Maximum daily reviews $r_{\max} = 37{,}293$, $y^* = 10.527$.	1996-05-20 to 2014-07-23 ($n = 6{,}639$).	SNAP Amazon product graph, Books ratings (S15).	Dataset article published 2015. Reviews span 1996–2014.

Continued on next page

Supplementary Table 1. Corpus composition and provenance *(continued)*.

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
bitcoin	Trade & Institutions	Cryptocurrency	OTC users and integer rating values.	User–rating–user incidences over the complete Bitcoin OTC history.	Rating timestamps.	Maximum daily ratings $r_{\max} = 319$, $y^* = 5.768$.	2010-11-08 to 2016-01-25 ($n = 1{,}905$).	PyTorch Geometric BitcoinOTC (S16).	Dataset article published 2016. Ratings span 2010–2016.
world	Trade & Institutions	International Development	World Bank lending instruments and countries.	Instrument–country eligibility.	Project board-approval dates.	Maximum daily project approvals $r_{\max} = 12$, $y^* = 2.565$.	2014-01-14 to 2024-12-20 ($n = 3{,}994$).	World Bank projects and country APIs (S17).	Mutable APIs retrieved 29 September 2026. Approval-year cutoff 2024.
wiki	Technology & Information	Knowledge Graphs	Wikipedia mathematics pages.	Page–page relations from WikiMaths snapshots.	Daily summed page activity.	Maximum daily summed page activity $r_{\max} = 1{,}323{,}743$, $y^* = 14.096$.	2019-03-16 to 2021-03-15 ($n = 731$).	UCI Wikipedia Math Essentials and PyTorch Geometric Temporal WikiMaths (S18).	UCI dataset released 2021. Activity spans 2019–2021.
jodie	Technology & Information	Interaction Networks	Wikipedia users and pages.	User–page interactions.	Interaction timestamps.	Maximum daily interactions $r_{\max} = 6{,}087$, $y^* = 8.714$.	Days 0–30 ($n = 31$).	PyTorch Geometric JODIE, Wikipedia (S14).	Dataset article published 2019. Timestamps reindexed to relative days.
overflow	Technology & Information	Online Forums	Stack Overflow users and calendar days.	Complete bipartite graph between users and days.	Comment, answer, and other timed interactions.	Maximum daily timed interactions $r_{\max} = 52{,}446$, $y^* = 10.868$.	2008-08-01 to 2016-03-06 ($n = 2{,}775$).	SNAP sx-stackoverflow (S19).	Dataset article published 2017. Interactions span 2008–2016.
email	Technology & Information	Email Exchanges	European research-institution email accounts and days.	Complete bipartite graph between accounts and days.	Email timestamps.	Maximum daily emails $r_{\max} = 2{,}270$, $y^* = 7.728$.	Days 0–803 ($n = 804$).	SNAP email-Eu-core-temporal (S19).	Dataset article published 2017. Timestamps reindexed to relative days.
college	Technology & Information	Social Networks	UC Irvine college-message users and days.	Complete bipartite graph between users and days.	Private-message timestamps.	Maximum daily private messages $r_{\max} = 2{,}678$, $y^* = 7.893$.	2004-04-15 to 2004-10-26 ($n = 195$).	SNAP CollegeMsg (S20).	Dataset article published 2009. Messages were recorded in 2004.
idling	Transportation & Infrastructure	Mobility	Halifax drive-network nodes from the unsimplified OSM graph.	Street segments (OSM drive graph, undirected).	Vehicle-idling records.	Maximum daily idling records $r_{\max} = 732$, $y^* = 6.597$.	2025-06-02 to 2025-09-02 ($n = 93$).	OpenStreetMap via OSMnx, query Halifax, Canada (S21, S22), and local idling CSVs (S29).	Data preprint first posted 2024 and revised 2025. Observations are from 2025. OSMnx article published 2025.

Continued on next page

Supplementary Table 1. Corpus composition and provenance *(continued)*.

System	Domain	Discipline	Nodes	Edges	Event	Target	Time (Days)	Source	Notes
windmill	Transportation & Infrastructure	Energy	Wind turbines.	Complete graph among turbines.	Raw-production on/off transitions (threshold 10^{-6}).	Maximum daily turbine transitions $r_{\max} = 7$, $y^* = 2.079$.	Days 0–727 ($n = 728$).	PyTorch Geometric Temporal WindmillOutput Large [87].	Loader article published 2021. Source supplies snapshot order rather than calendar dates.
metrla	Transportation & Infrastructure	Traffic	METR-LA loop detectors.	Detector adjacency from METR-LA snapshots.	Onset of congestion (raw speed at or below the detector-wise 1st percentile).	Maximum daily congestion onsets $r_{\max} = 675$, $y^* = 6.516$.	2012-03-01 to 2012-06-27 ($n = 119$).	PyTorch Geometric Temporal METRLA [87] (S23).	Benchmark article published 2018 and loader article 2021. Sensor observations are from 2012.
pemsbay	Transportation & Infrastructure	Traffic II	PEMS-BAY loop detectors.	Detector adjacency from PEMS-BAY snapshots.	Onset of congestion (raw speed at or below the detector-wise 1st percentile).	Maximum daily congestion onsets $r_{\max} = 634$, $y^* = 6.454$.	2017-01-01 to 2017-06-29 ($n = 180$).	PyTorch Geometric Temporal PEMS-BAY [87] (S23).	Benchmark article published 2018 and loader article 2021. Sensor observations are from 2017.
montevideo	Transportation & Infrastructure	Ridership	Montevideo bus stops.	Stop adjacency from MontevideoBus snapshots.	Stop-hours with standardized inflow above the global 99th percentile.	Maximum daily exceedance stop-hours $r_{\max} = 256$, $y^* = 5.549$.	Days 0–29 ($n = 30$).	PyTorch Geometric Temporal MontevideoBus [87] (S28).	Data recorded in October 2020. Loader article published 2021. Dates represented as relative days.

Each row represents one N3S, with observations aggregated to r_t (Eq. (2)) and targets $r_{\max}$ and $y^* = \ln(1 + r_{\max})$ (Eq. (3)). Metrla, pemsbay, montevideo, windmill, and wiki provide complete daily series, including zero-event days. Chickenpox provides weekly county totals, summed and divided by 7, with one daily-average value per report date; the other 24 systems use daily event counts. The remaining 19 series include only days with recorded events. Windows follow native recording periods, expressed as dates or relative day indices; n is the inclusive span in days, not necessarily the series length L. The 7 API systems (gwosc, river, seismic, rain, faers, federal, and world) also use fixed script query dates, with end cutoffs in the source notes. Durations were not standardized, and maxima remain conditional on their windows. Observation dates differ from source release or publication dates; source notes specify versions and relative-date conventions. Additional processing conventions are provided in the supplement (Supplementary Note 2). Citations S1–S31 refer to the supplementary sources (Supplementary References).

Supplementary Table 2. Domain transfer and baseline frontier performance.

Evaluation	Method	Median EI [Q1–Q3]	VR	MV	MS
Transfer	Domain (LOGO-CV)	0.75 [0.74–0.77]	0.04	0.08	6.86
	Discipline (LOGO-CV)	0.73 [0.72–0.76]	0.00	0.00	6.12
Baseline	Random (5-Fold-CV)	0.77 [0.76–0.78]	0.02	0.06	6.86
	Random (10-Fold-CV)	0.76 [0.75–0.78]	0.01	0.02	6.87

Entries are medians of estimator-level means over folds or groups and 30 repeats (seeds 42–71), with quartiles across 9 estimators in brackets. Groups, folds, repeats, and estimators are equally weighted. LOGO-CV folds are fixed across repeats; random folds are reshuffled. EI denotes the Efficiency Index (Performance Metrics, Eq. (8)), VR the violation rate, MV the mean exceedance among within-fold violators (0 if none), and MS the mean non-negative slack across evaluated systems. MV and MS use natural-log target units. Descriptive summaries only.

Supplementary Table 3. Equivalence of resampling regimes to domain-level LOGO-CV.

Domain Transfer Equivalence						
Evaluation	Median ΔEI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Discipline (LOGO-CV)	0.00	-0.08	< 0.001	< 0.001	***	Yes
Random (5-Fold-CV)	0.00	0.40	< 0.001	< 0.001	***	Yes
Random (10-Fold-CV)	0.00	0.24	< 0.001	< 0.001	***	Yes
Estimation-Observation Consensus Equivalence						
Evaluation	Median ΔCI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Discipline (LOGO-CV)	0.00	-0.14	< 0.001	< 0.001	***	Yes
Random (5-Fold-CV)	-0.01	-0.32	< 0.001	< 0.001	***	Yes
Random (10-Fold-CV)	0.00	-0.34	< 0.001	< 0.001	***	Yes

Each contrast uses 45 paired estimator-by-domain cells (9 estimators $\times$ 5 domains). Out-of-sample predictions were averaged over 30 seeds (42–71) before computing EI and CI within the same 5 broad domains for every regime. EI and CI are defined in the methodology (Performance Metrics, Eqs. (8) and (9)). Shifts are medians of paired differences (test comparison $-$ domain-level LOGO-CV), not differences of marginal medians. Rank-biserial r uses nonzero, unshifted differences with average ranks for ties; positive values favor the comparison regime. TOST uses paired one-sided Wilcoxon signed-rank tests on $\Delta + \delta$ and $\Delta - \delta$, assuming a symmetric paired-difference distribution with location θ. The nulls are $\theta \leq -\delta$ and $\theta \geq \delta$; unadjusted p is the larger of the two one-sided values. Margins $\delta_{\mathrm{EI}} = 0.02$ and $\delta_{\mathrm{CI}} = 0.11$ are IQRs across the 9 reference estimator-level means after equal averaging across domains, floored to 2 decimals. Raw differences with magnitude $< 10^{-10}$ were zeroed before shifting. Tests used `SciPy 1.16.3`, `zero_method='wilcox'`, `method='auto'`, and `correction=False`, discarding zeros after shifting. p_{H} is Holm-adjusted across the 3 regime comparisons separately for EI and CI. Eq. requires rejection of both nulls at $p_{\mathrm{H}} < 0.05$. Shared estimators, training systems, and domains make inference conditional on this dependent panel (Evaluation and Statistical Inference).

Supplementary Table 4. Perturbation effects under the highest test intensity.

Perturbation	Method	Median ΔEI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Domain Transfer							
Network	Sample	0.00	-0.35	< 0.001	< 0.001	***	Yes
	Rewire	0.00	-0.23	< 0.001	< 0.001	***	Yes
	Densify	0.00	-0.40	< 0.001	< 0.001	***	Yes
Invariants	Jitter	0.00	-0.45	< 0.001	< 0.001	***	Yes
	Noise	0.00	-0.60	< 0.001	< 0.001	***	Yes
	Subset	0.00	0.05	< 0.001	< 0.001	***	Yes
Process	Bootstrapping	0.01	0.33	< 0.001	< 0.001	***	Yes
	Scaling	-0.01	-0.45	< 0.001	< 0.001	***	Yes
	Smoothing	-0.02	-0.75	< 0.001	< 0.001	***	Yes
Signatures	Jitter	0.00	-0.31	< 0.001	< 0.001	***	Yes
	Noise	0.00	-0.44	< 0.001	< 0.001	***	Yes
	Subset	-0.01	-0.61	< 0.001	< 0.001	***	Yes
Estimation-Observation Consensus							
Perturbation	Method	Median ΔCI	Rank-biserial r	TOST p	Holm-adj. p	Sig.	Eq.
Network	Sample	0.00	-0.58	< 0.001	< 0.001	***	Yes
	Rewire	0.00	-0.43	< 0.001	< 0.001	***	Yes
	Densify	0.00	-0.47	< 0.001	< 0.001	***	Yes
Invariants	Jitter	0.00	-0.44	< 0.001	< 0.001	***	Yes
	Noise	0.00	-0.42	< 0.001	< 0.001	***	Yes
	Subset	0.00	-0.41	< 0.001	< 0.001	***	Yes
Process	Bootstrapping	-0.11	-0.82	0.028	0.028	*	Yes
	Scaling	-0.02	-0.23	< 0.001	< 0.001	***	Yes
	Smoothing	-0.06	-0.73	0.002	0.003	**	Yes
Signatures	Jitter	-0.01	-0.29	< 0.001	< 0.001	***	Yes
	Noise	-0.01	-0.68	< 0.001	< 0.001	***	Yes
	Subset	-0.04	-0.82	< 0.001	< 0.001	***	Yes

Frozen domain-level LOGO-CV retains original-data training fits and uses 45 paired estimator-by-domain cells (9 estimators × 5 domains), summarized over 30 seeds (42–71). EI is averaged over repetitions in both conditions. CI is computed from seed-averaged predictions for the original and deterministic conditions, but averaged over paired seed–realization metrics for stochastic perturbations. EI and CI are defined in the methodology (Performance Metrics, Eqs. (8) and (9)). Thus aggregation order can contribute to stochastic CI shifts. Shifts are medians of paired differences (perturbed − original). Rank-biserial r uses nonzero, unshifted differences with average ranks for ties; positive values favor perturbation. Only each method's most severe tested setting is included (Supplementary Table 9); Sample denotes node removal and coordinate subsetting. Original maximum targets are retained. TOST uses paired one-sided Wilcoxon signed-rank tests on $\Delta + \delta$ and $\Delta - \delta$, assuming a symmetric paired-difference distribution with location θ. The nulls are $\theta \leq -\delta$ and $\theta \geq \delta$; unadjusted p is the larger of the two one-sided values. Margins $\delta_{\mathrm{EI}} = 0.08$ and $\delta_{\mathrm{CI}} = 0.14$ are IQRs across the 45 original-condition cells, floored to 2 decimals. Raw differences with magnitude $< 10^{-10}$ were zeroed before shifting. Tests used `SciPy 1.16.3`, `zero_method='wilcox'`, `method='auto'`, and `correction=False`, discarding zeros after shifting. p_{H} is Holm-adjusted across the 12 methods separately for EI and CI. Eq. requires rejection of both nulls at $p_{\mathrm{H}} < 0.05$. Shared estimators, training systems, and domains make inference conditional on this dependent panel (Evaluation and Statistical Inference).

Supplementary Table 5. Falsification effects under frozen and retrained frontiers.

Falsification	Method	Median ΔEI	Rank-biserial r	One-sided p	Holm-adj. p	Sig.	Diff.
Domain Transfer							
Frozen	Target Permutation	-0.02	-0.38	0.013	0.013	*	Yes
	Random Generation	-0.04	-0.91	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.03	-0.74	< 0.001	< 0.001	***	Yes
Retrain	Target Permutation	-0.09	-0.53	< 0.001	0.001	**	Yes
	Random Generation	-0.02	-0.59	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.03	-0.57	< 0.001	< 0.001	***	Yes
Estimation-Observation Consensus							
Falsification	Method	Median ΔCI	Rank-biserial r	One-sided p	Holm-adj. p	Sig.	Diff.
Frozen	Target Permutation	-0.29	-0.97	< 0.001	< 0.001	***	Yes
	Random Generation	-0.28	-0.91	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.21	-0.88	< 0.001	< 0.001	***	Yes
Retrain	Target Permutation	-0.24	-0.99	< 0.001	< 0.001	***	Yes
	Random Generation	-0.18	-0.94	< 0.001	< 0.001	***	Yes
	Vector Generation	-0.31	-0.97	< 0.001	< 0.001	***	Yes

Each domain-level LOGO-CV contrast uses 45 paired estimator-by-domain cells (9 estimators × 5 domains). Predictions are averaged over 30 model-fitting seeds (42–71) before computing EI and CI within each domain. EI and CI are defined in the methodology (Performance Metrics, Eqs. (8) and (9)). Each falsified corpus was generated once with seed 42; model-fitting repeats are not independent negative-control corpora. Frozen frontiers retain original-data training fits, while retrained frontiers refit standardizers and estimators on the falsified training folds. Shifts are medians of paired differences (falsified − original), with negative values favoring the original condition. Rank-biserial r uses nonzero, unshifted differences with average ranks for ties. One-sided Wilcoxon signed-rank tests assess $H_0 : \theta \geq 0$ versus $H_1 : \theta < 0$, assuming a symmetric paired-difference distribution with location θ. Raw differences with magnitude $< 10^{-10}$ were zeroed. Tests used `SciPy 1.16.3`, `zero_method='wilcox'`, `method='auto'`, and `correction=False`, discarding zero differences. p is unadjusted; p_{H} is Holm-adjusted across the 6 method-by-frontier comparisons separately for EI and CI. Diff. requires $p_{\mathrm{H}} < 0.05$ and a negative median shift, supporting inferiority to the original condition. Shared estimators, training systems, and domains make inference conditional on this dependent panel (Evaluation and Statistical Inference).

Supplementary Table 6. Non-inferiority of the log-additive decomposition to alternative formulations.

Ablation	Method	Median ΔEI	Rank-biserial r	One-sided p	Holm-adj. p	Sig.	NI.
Domain Transfer							
Simple	Invariants Only	-0.01	-0.77	< 0.001	< 0.001	***	Yes
	Signatures Only	-0.01	-0.21	< 0.001	< 0.001	***	Yes
Complex	Joint Features	-0.02	-0.55	< 0.001	< 0.001	***	Yes
	Interaction Terms	-0.01	-0.46	< 0.001	< 0.001	***	Yes
Estimation-Observation Consensus							
Ablation	Method	Median ΔCI	Rank-biserial r	One-sided p	Holm-adj. p	Sig.	NI.
Simple	Invariants Only	-0.20	-0.86	< 0.001	< 0.001	***	Yes
	Signatures Only	-0.08	-0.55	< 0.001	< 0.001	***	Yes
Complex	Joint Features	-0.27	-0.67	< 0.001	< 0.001	***	Yes
	Interaction Terms	-0.10	-0.65	< 0.001	< 0.001	***	Yes

Each retrained domain-level LOGO-CV contrast uses 45 paired estimator-by-domain cells (9 estimators × 5 domains) and 30 model-fitting seeds (42–71), with all pipelines refit within their training folds. EI is averaged over repetitions; CI is computed from seed-averaged out-of-sample predictions within each domain. EI and CI are defined in the methodology (Performance Metrics, Eqs. (8) and (9)). Shifts are medians of paired differences (alternative formulation − log-additive decomposition); negative values favor the decomposition. Rank-biserial r uses nonzero, unshifted differences with average ranks for ties, independently of the tested margin. One-sided Wilcoxon signed-rank tests assess $H_0 : \theta \geq \delta$ versus $H_1 : \theta < \delta$, assuming a symmetric paired-difference distribution with location θ. Margins $\delta_{\mathrm{EI}} = 0.08$ and $\delta_{\mathrm{CI}} = 0.14$ are IQRs across the 45 log-additive reference cells, floored to 2 decimals. Raw differences with magnitude $< 10^{-10}$ were zeroed before subtracting δ. Tests used `SciPy 1.16.3`, `zero_method='wilcox'`, `method='auto'`, and `correction=False`, discarding zeros after shifting. p is unadjusted; p_{H} is Holm-adjusted separately within the 2 simple and 2 complex alternatives for EI and CI. NI. requires $p_{\mathrm{H}} < 0.05$ and a median shift below the margin. This supports non-inferiority within the stated margin, not superiority over the alternatives. Shared estimators, training systems, and domains make inference conditional on this dependent panel (Evaluation and Statistical Inference).

Supplementary Table 7. Graph invariant definitions, expressions, and asymptotic time complexity.

Category	Invariant	Definition	Expression	Complexity
Simple	1. Number of Nodes	Number of nodes	$n = \|V\|$	$O(1)$
	2. Number of Edges	Number of edges	$m = \|E\|$	$O(1)$
	3. Articulation Points	Vertices whose removal splits G	$\|\{v : \omega(G-v) > \omega(G)\}\|$	$O(n+m)$
	4. Bridges	Edges whose removal splits G	$\|\{e : \omega(G-e) > \omega(G)\}\|$	$O(n+m)$
Cohesion	5. Diameter	Longest shortest-path length †	$\max_v \epsilon(v)$	$O(nm)$
	6. Radius	Smallest vertex eccentricity †	$\min_v \epsilon(v)$	$O(nm)$
	7. Degeneracy	Largest k with a k-core	$k^* = \max_v c(v)$	$O(n+m)$
	8. k-Core Size	Vertices in the core at k^*	$\|\{v : c(v) = k^*\}\|$	$O(n+m)$
Extremal	9. Maximum Degree	Highest vertex degree	$\max_v d_v$	$O(n)$
Statistical	10. Degree Variance	Spread of vertex degrees	$n^{-1}\sum_v (d_v - \bar{d})^2$	$O(n)$
	11. Degree Entropy	Entropy of degree distribution	$-\sum_k p_k \ln p_k$	$O(n)$
	12. Joint Degree Entropy	Entropy of edge degree pairs	$-\sum_{k\le l} q_{kl} \ln q_{kl}$	$O(m)$
	13. Degree Skewness	Asymmetry of degree distribution	$\frac{\sqrt{n(n-1)}}{n-2}\, g_1$	$O(n)$
	14. Degree Kurtosis	Tail weight of degree distribution	$\frac{(n-1)[(n+1)g_2+6]}{(n-2)(n-3)}$	$O(n)$
	15. Global Clustering Coefficient	Fraction of closed triples	$3N_\triangle / N_\wedge$	$O(nd_{\max}^2)$
	16. Assortativity	Correlation of endpoint degrees	$\mathrm{corr}_{\{u,v\}\in E}(d_u, d_v)$	$O(m)$
Spectral	17. Normalized Laplacian 2nd Moment	Diffusion intensity	$\mathrm{tr}(L^2)/n$	$O(m)$
	18. Normalized Laplacian 3rd Moment	Diffusion asymmetry	$\mathrm{tr}(L^3)/n$	$O(m^2)$
	19. Random Walk 3rd Moment	Three-step recurrence	$\mathrm{tr}(P^3)/n$	$O(m^2)$
	20. Random Walk 4th Moment	Four-step recurrence	$\mathrm{tr}(P^4)/n$	$O(m^2)$
	21. Adjacency 4th Moment	Four-length walk density	$\mathrm{tr}(A^4)/n$	$O(m^2)$

Admission required global graph descriptors defined solely by topology, independent of embeddings, weights, external parameters, and arbitrary thresholds, exactly and deterministically computable at corpus scale, and non-trivially variable across systems. All returned values were finite. † Diameter and radius use the largest connected component; degeneracy and k-core size use the full graph. Spectral moments use exact trace identities without eigendecomposition; the 4 complete bipartite substrates use closed-form invariants. Here $n = |V|$, $m = |E|$, d_v is vertex degree, $\bar{d}$ its mean, $\mu_j = n^{-1}\sum_v (d_v - \bar{d})^j$, $g_1 = \mu_3/\mu_2^{3/2}$, and $g_2 = \mu_4/\mu_2^2 - 3$. ω counts connected components, $\epsilon(v)$ is eccentricity in hops, $c(v)$ is core number, p_k is the fraction of nodes of degree k, q_{kl} is the fraction of edges with unordered endpoint degrees $\{k, l\}$, and $N_\triangle$ and $N_\wedge$ count triangles and connected triples. A is the adjacency matrix, $\hat{A} = D^{-1/2}AD^{-1/2}$ over non-isolated nodes, $L = I - \hat{A}$, and $P = D^{-1}A$. Isolated nodes count toward n but contribute 0 to $\mathrm{tr}(L^k)$. Logarithms are natural; assortativity counts both edge orientations. Undefined assortativity and clustering are 0. Degree skewness and kurtosis are 0 for equal degrees or $n < 3$ and $n < 4$, respectively. References: [100–110].

Supplementary Table 8. Process signature definitions, expressions, and asymptotic time complexity.

Property	Signature	Definition	Expression	Complexity
Dependence	1. Lag-1 Autocorrelation	Short-term sequential dependence	$\frac{\sum_{i=1}^{L-1} \tilde{r}_i \tilde{r}_{i+1}}{\sum_{i=1}^{L} \tilde{r}_i^2}$	$O(L)$
	2. Hurst Exponent	Long-range dependence	$\ln(R/s)/\ln L$	$O(L)$
Dispersion	3. Coefficient of Variation	Scale-free dispersion	$s/\bar{r}$	$O(L)$
	4. Fano Factor	Deviation from Poisson dispersion	$s^2/\bar{r}$	$O(L)$
Volatility	5. Normalized Successive Difference	Local temporal volatility	$\frac{1}{(L-1)\,\bar{r}} \sum_{i=1}^{L-1} \lvert r_{i+1} - r_i \rvert$	$O(L)$
Trend	6. Trend Coefficient	Systematic non-stationarity	$\mathrm{corr}(r_i, i)$	$O(L)$
Asymmetry	7. Count Skewness	Distributional asymmetry	$\frac{1}{L} \sum_{i=1}^{L} (\tilde{r}_i/s)^3$	$O(L)$

Admission required global series descriptors defined solely by the rate series, without external parameters or arbitrary thresholds, exactly and deterministically computable, and non-trivially variable across systems. All returned values were finite. For the chronologically ordered rate series $r_1, \ldots, r_L$ (Eq. (2)), $L = |T|$, $\bar{r}$ is the mean, $\tilde{r}_i = r_i - \bar{r}$, and s is the sample standard deviation with denominator $L - 1$. Interval inclusion follows corpus definitions (Supplementary Table 1); lag-1 dependence and trend index consecutive included observations, not calendar gaps. The Hurst exponent is a full-series, single-window rescaled-range estimate clipped to $[0, 1]$, with $R = \max_i Y_i - \min_i Y_i$ and cumulative sums $Y_i = \sum_{j \le i} \tilde{r}_j$. Trend is Pearson correlation with position i. CV, Fano factor, and normalized successive difference use denominator $\bar{r} + 10^{-10}$ and are 0 when $\bar{r} \le 0$. Lag-1 autocorrelation and trend are 0 for denominators $\le 10^{-10}$. Skewness is 0 when $L < 3$ or $s \le 10^{-10}$; Hurst is 0.5 when $L < 8$ or R or s is $\le 10^{-10}$. References: [111–119].

Supplementary Table 9. Stress-test protocols and inference.

Category	Method	Description	Frontier
Perturbation (TOST equivalence at maximum intensity, $\delta_{\mathrm{EI}} = 0.08$, $\delta_{\mathrm{CI}} = 0.14$)			
Network	Rewire	Degree-preserving edge swaps, fraction 0.05–0.35 (max 0.35)	Frozen
	Densify	Uniform edge addition, fraction 0.05–0.35 (max 0.35)	Frozen
	Sample	Uniform node removal, fraction 0.05–0.35 (max 0.35)	Frozen
Invariants	Noise	Additive Gaussian noise, corpus-wide, 0.05–0.35 (max 0.35)	Frozen
	Jitter	Multiplicative jitter, 0.05–0.35 (max 0.35)	Frozen
	Subset	Random coordinate subset, retention 0.95–0.65 (max 0.65)	Frozen
Process	Bootstrapping	Bootstrap to destroy temporal order 0.05–0.35 (max 0.05)	Frozen
	Scaling	Power-law scaling, exponent 0.25–1.75 (max 0.25)	Frozen
	Smoothing	Rolling-mean smoothing, window 3–15 intervals (max 15)	Frozen
Signatures	Noise	Additive Gaussian noise, corpus-wide, 0.05–0.35 (max 0.35)	Frozen
	Jitter	Multiplicative jitter, 0.05–0.35 (max 0.35)	Frozen
	Subset	Random coordinate subset, retention 0.95–0.65 (max 0.65)	Frozen
Falsification (one-sided inferiority to the original condition, Holm-adjusted)			
Negative	Target Permutation	Cross-domain derangement of y^* with x and z unchanged	Frozen + Retrain
	Random Generation	Each feature redrawn uniformly within its observed corpus range	Frozen + Retrain
	Vector Generation	Feature bootstrap plus Gaussian noise at pooled dispersion	Frozen + Retrain
Ablation (one-sided non-inferiority of the log-additive decomposition, $\delta_{\mathrm{EI}} = 0.08$, $\delta_{\mathrm{CI}} = 0.14$)			
Simple	Invariants Only	Frontier fitted on x' alone	Retrain
	Signatures Only	Frontier fitted on z' alone	Retrain
Complex	Joint Features	x' and z' concatenated into a single-stage fit	Retrain
	Interaction Terms	Augmented with all 147 pairwise products	Retrain

Stress tests use 30 model-fitting seeds (42–71), paired one-to-one with 30 independent stochastic perturbation realizations or applied to a single deterministic realization. Grids contain 7 equally spaced values between the listed endpoints; scaling includes the identity $\gamma = 1$. Feature interventions act on raw coordinates before fold-wise standardization. Additive-noise standard deviation is $s\sigma_j$, where σ_j is the sample standard deviation of coordinate j across all 25 original systems. Masking retains $\lfloor \rho p \rfloor$ coordinates, with $p = 21$ for invariants and $p = 7$ for signatures; noise, jitter draws, and masks are shared across systems within a realization. Network and process interventions recompute the affected feature vector while retaining original maximum targets. Complete bipartite node removal uses exact closed-form invariants; rewiring and densification use deterministic approximations (Supplementary Note 1), with rewiring interpolated toward configuration-model estimates [120]. Rewiring and densification leave the complete graphs gwosc, seismic, and windmill unchanged. Falsified corpora are generated once with seed 42 as corpus-level negative controls. "Max" denotes the most severe setting, not necessarily the largest parameter: severity is intensity ϵ for network methods, intensity s for noise and jitter, window w for smoothing, $1 - \rho$ for retention ρ, $1 - s$ for bootstrapping, and $|\ln \gamma|$ for scaling exponent γ. Inference uses the 45 paired estimator-by-domain cells (9 estimators × 5 domains); metric aggregation, margins, and Holm-adjustment families are specified across test summaries (Supplementary Tables 4–6).

Supplementary Note 1: Complete Bipartite Perturbations

The 4 complete bipartite substrates (college, email, mooc, overflow) were represented by their fold sizes m and n, with $N = m + n$ nodes and $M = mn$ edges, because their explicit graphs reach up to 7,200,000,000 edges. Node removal at intensity ϵ deleted $\lfloor \epsilon N \rfloor$ nodes drawn uniformly without replacement, counted the removed nodes in each fold, and evaluated the closed forms (Supplementary Table 7) for the remaining complete bipartite graph. This computation is exact and used 30 realizations. Rewiring and densification used a deterministic degree-class approximation with 1 realization.

For degree classes c with degree d_c and N_c nodes, and an edge count M', let

$$N = \sum_c N_c, \quad S_1 = \sum_c N_c d_c, \quad S_2 = \sum_c N_c d_c^2, \quad \bar{d} = \frac{S_1}{N}, \quad \kappa = \frac{S_2}{\max(S_1, 1)} - 1, \quad \overline{d^{-1}} = \frac{1}{N} \sum_c \frac{N_c}{\max(d_c, 1)}.$$

The approximation Φ assigned the following values and set articulation points, bridges, clustering, and assortativity to 0.

- Size: $|V| = N$, $|E| = M'$, and $d_{\max} = \max_c d_c$.
- Cohesion: diameter $D = \max\{2, \ln \max(N, 2) / \ln \kappa\}$ when $\kappa > 1$ and $D = N$ otherwise, radius $D/2$, degeneracy $k^* = \min(d_{\max}, \sqrt{2M'})$, and k-core size $\sum_c N_c \mathbf{1}[d_c \geq k^*]$.
- Degree statistics: variance $N^{-1} \sum_c N_c (d_c - \bar{d})^2$, bias-corrected skewness and excess kurtosis of the class-weighted degrees, degree entropy $-\sum_k p_k \ln p_k$ over degrees rounded to integers k with vertex fractions p_k, and joint degree entropy $-2 \sum_k q_k \ln q_k$ with stub fractions $q_k \propto k p_k$.
- Spectral moments: $\mathrm{tr}(L^2)/N = 1 + 2M'N / \max(S_1, 1)^2$, $\mathrm{tr}(L^3)/N = 1 + (S_2 - S_1)^3 / \max(N S_1^3, 1)$, $\mathrm{tr}(P^3)/N = 6(S_2 - S_1)^2 \, \overline{d^{-1}} / [\max(6S_1, 1) \max(S_1, 1)]$, $\mathrm{tr}(P^4)/N = N^{-1} \sum_c N_c [d_c S_2 / \max(S_1^2, 1) + \max(\kappa, 0) / \max(d_c, 1)]$, and $\mathrm{tr}(A^4)/N = [S_2 + S_2^2 / \max(2S_1, 1)]/N$.

Rewiring at intensity ϵ evaluated Φ on the classes of $K_{m,n}$, with degrees (n, m), class sizes (m, n), and $M' = M$, and interpolated each invariant as $(1 - \alpha) x_j + \alpha \Phi_j$ with $\alpha = 1 - (1 - \epsilon)^2$, where x_j is the exact $K_{m,n}$ value. Densification added $a = \min(\lfloor \epsilon M \rfloor, C)$ edges among the $C = \binom{m}{2} + \binom{n}{2}$ within-fold vertex pairs, so that each pair was added with probability $q = a/C$. It evaluated Φ with degrees $(n + q(m-1), m + q(n-1))$, class sizes (m, n), and $M' = M + a$. It then multiplied the $K_{m,n}$ articulation points, bridges, and assortativity by $1 - q$ and set clustering to $\min\{c_0 + (1 - c_0) q, 1\}$, where c_0 is the $K_{m,n}$ clustering. All other substrates were perturbed as explicit graphs.

Supplementary Note 2: Corpus Processing Conventions

The **federal** series excludes awards with non-positive amounts or start dates outside 2011-01-01 to 2024-12-31. Zero speeds are treated as missing in **metrla** (8.1% of readings) and as readings in **pemsbay** (0.003%). **Montevideo** drops the first 4 hours removed by the loader lag and the final 20 hours of the 744-hour source that do not complete a day.

GPS timestamps in **gwosc** and **auger** were converted to UTC without leap-second correction; correction would move 19 **auger** events and no **gwosc** events to another day. The **crop** network retains 12 nodes and 30 edges from studies whose visits lack dates. Exact-name matching of **world** countries to lending metadata leaves 23 regional aggregates and spelling variants isolated.

Supplementary References

S1. Vallisneri, M., Kanner, J., Williams, R., Weinstein, A. & Stephens, B. The LIGO Open Science Center. *J. Phys. Conf. Ser.* **610**, 012021 (2015). https://doi.org/10.1088/1742-6596/610/1/012021

S2. U.S. Geological Survey. USGS Water Data for the Nation (1994). https://doi.org/10.5066/F7P55KJN

S3. Pierre Auger Collaboration. Pierre Auger Observatory Open Data, version 3 [Dataset]. Zenodo (2024). https://doi.org/10.5281/zenodo.10488964

S4. Albuquerque Seismological Laboratory/USGS. Global Seismograph Network (GSN - IRIS/USGS) [Dataset]. International Federation of Digital Seismograph Networks (1988). https://doi.org/10.7914/SN/IU

S5. U.S. Geological Survey. Advanced National Seismic System (ANSS) Comprehensive Catalog (2017). https://doi.org/10.5066/F7MS3QZH

S6. Meteostat. Meteostat weather and climate database. https://meteostat.net (accessed 1 September 2026).

S7. Cassada, R. C. & Russell, R. L. The dauerlarva, a post-embryonic developmental variant of the nematode *Caenorhabditis elegans*. *Dev. Biol.* **46**, 326–342 (1975). https://doi.org/10.1016/0012-1606(75)90109-8

S8. Golden, J. W. & Riddle, D. L. The *Caenorhabditis elegans* dauer larva: developmental effects of pheromone, food, and temperature. *Dev. Biol.* **102**, 368–378 (1984). https://doi.org/10.1016/0012-1606(84)90201-X

S9. Allen-Perkins, A. et al. CropPol: A dynamic, open and global database on crop pollination. *Ecology* **103**, e3614 (2022). https://doi.org/10.1002/ecy.3614

S10. Hungarian Chickenpox Cases [Dataset]. UCI Machine Learning Repository (2021). https://doi.org/10.24432/C5103B

S11. U.S. Food and Drug Administration. openFDA Drug Event API. https://open.fda.gov/apis/drug/event/ (accessed 13 March 2026).

S12. Guttag, J. CHB-MIT Scalp EEG Database, version 1.0.0 [Dataset]. PhysioNet (2010). https://doi.org/10.13026/C2K01R

S13. U.S. Department of the Treasury. USAspending API. https://api.usaspending.gov/ (accessed 25 August 2026).

S14. Kumar, S., Zhang, X. & Leskovec, J. Predicting dynamic embedding trajectory in temporal interaction networks. In *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining* 1269–1278 (ACM, 2019). https://doi.org/10.1145/3292500.3330895

S15. McAuley, J., Targett, C., Shi, Q. & van den Hengel, A. Image-based recommendations on styles and substitutes. In *Proc. 38th Int. ACM SIGIR Conf. Research and Development in Information Retrieval* 43–52 (ACM, 2015). https://doi.org/10.1145/2766462.2767755

S16. Kumar, S., Spezzano, F., Subrahmanian, V. S. & Faloutsos, C. Edge weight prediction in weighted signed networks. In *Proc. IEEE 16th Int. Conf. Data Mining* 221–230 (IEEE, 2016). https://doi.org/10.1109/ICDM.2016.0033

S17. World Bank. Projects & Operations API and Country API. https://search.worldbank.org/api/v2/projects and https://api.worldbank.org/v2/country (accessed 29 September 2026).

S18. Wikipedia Math Essentials [Dataset]. UCI Machine Learning Repository (2021). https://doi.org/10.24432/C5V32F

S19. Paranjape, A., Benson, A. R. & Leskovec, J. Motifs in temporal networks. In *Proc. 10th ACM Int. Conf. Web Search and Data Mining* 601–610 (ACM, 2017). https://doi.org/10.1145/3018661.3018731

S20. Panzarasa, P., Opsahl, T. & Carley, K. M. Patterns and dynamics of users' behavior and interaction: network analysis of an online community. *J. Am. Soc. Inf. Sci. Technol.* **60**, 911–932 (2009). https://doi.org/10.1002/asi.21015

S21. OpenStreetMap contributors. OpenStreetMap [Dataset]. https://www.openstreetmap.org (accessed 10 November 2025).

S22. Boeing, G. Modeling and analyzing urban networks and amenities with OSMnx. *Geogr. Anal.* **57**, 567–577 (2025). https://doi.org/10.1111/gean.70009

S23. Li, Y., Yu, R., Shahabi, C. & Liu, Y. Diffusion convolutional recurrent neural network: data-driven traffic forecasting. In *Int. Conf. Learning Representations* (2018).

S24. Abbott, B. P. et al. GWTC-1: A Gravitational-Wave Transient Catalog of Compact Binary Mergers Observed by LIGO and Virgo during the First and Second Observing Runs. *Phys. Rev. X* **9**, 031040 (2019). https://doi.org/10.1103/PhysRevX.9.031040

S25. Abbott, R. et al. GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo during the First Half of the Third Observing Run. *Phys. Rev. X* **11**, 021053 (2021). https://doi.org/10.1103/PhysRevX.11.021053

S26. Abbott, R. et al. GWTC-2.1: Deep extended catalog of compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run. *Phys. Rev. D* **109**, 022001 (2024). https://doi.org/10.1103/PhysRevD.109.022001

S27. Abbott, R. et al. GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo during the Second Part of the Third Observing Run. *Phys. Rev. X* **13**, 041039 (2023). https://doi.org/10.1103/PhysRevX.13.041039

S28. Government of Uruguay. National Open Data Catalog. https://catalogodatos.gub.uy/ (accessed 1 September 2026).

S29. Kunz, N. & Gao, H. O. Global geolocated realtime data of interfleet urban transit bus idling. Preprint at https://doi.org/10.48550/arXiv.2403.03489 (2024).

S30. LIGO Scientific Collaboration, Virgo Collaboration & KAGRA Collaboration. GWTC-4.0 Data Release Documentation [Dataset]. Gravitational Wave Open Science Center (2025). https://doi.org/10.7935/aes8-px89

S31. Rader, R. et al. Non-bee insects are important contributors to global crop pollination. *Proc. Natl Acad. Sci. USA* **113**, 146–151 (2016). https://doi.org/10.1073/pnas.1517092112